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COMPREHENSIVE REVIEWER

LET-TOS PPST-Based

SPECIALIZATION

MATHEMATICS

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MATHEMATICS

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Arithmetic Number Theory and Business Math

1. Simplify expressions involving series of operations

1. Perform the indicated operations: $2[3+(-1)-4]$
   A. -2
   B. -3
   C. -4
   D. -5

2. Simplify: $\frac{1}{2}[5-(-1)]+(-6)$
   A. -1
   B. -3
   C. -5
   D. -7

3. Perform the indicated operations: $2^3+(-7+11)^2$
   A. 14
   B. 24
   C. 34
   D. 44

4. Simplify: $\frac{-9+3^2}{2^2-1}$
   A. 7
   B. 4
   C. 3
   D. 0

For numbers 5-9, consider the functions below:

$$f(x) = 3x^2 + 4x - 5$$
$$g(x) = 3x + 9$$
$$h(x) = x + 3$$

5. What is $(f+g)(x)$?
   A. $3x^2 + 7x - 4$
   B. $3x^2 + 7x + 4$
   C. $3x^2 - 7x - 4$
   D. $3x^2 - 7x + 4$

6. What is $(h-f)(x)$?
   A. $3x^2 - 3x + 8$
   B. $-3x^2 + 3x + 8$
   C. $3x^2 - 3x + 8$
   D. $-3x^2 - 3x + 8$

7. What is $\left(\frac{h}{g}\right)(x)$?
   A. $\frac{1}{3}$
   B. 3
   C. $x + 3$
   D. $3x + 9$

8. Evaluate: $f(0) + g(-1)$

A. 1
B. 4
C. 5
D. 8

9. Evaluate: $[g(2)][h(-1)]$

A. 15
B. 30
C. 45
D. 60

10. What is the sum of $x + 7$and $-4x - 5$?

A. $x + 3$
B. $2x + 8$
C. $-5x + 12$
D. $-3x + 2$

11. Simplify: $(2x + 1)(x - 4)$

A. $2x^2$
B. $x^2 - 7x$
C. $2x^2 - 7x - 4$
D. $x^2 - 7x - 4$

12. Simplify: $(5x^4 + 2x^3 - x^2 + 3x - 4) + (-2x^3 - 3x + 4)$

A. $5x^4 - x^2$
B. $5x^4 - x^2 + 8$
C. $5x^4 - x^2 + 6x$
D. $5x^4 + 4x^3 - x^2$

13. What is the remainder of $x^2 + 4x - 5$ divided by $x + 2$?

A. $x + 2$
B. $\frac{9}{x+2}$
C. $-\frac{9}{x+2}$
D. -1

14. What is the quotient of $15x^2 - 17x - 4$ divided by $5x + 1$?

A. 3
B. $3x$
C. $-3x$
D. $3x - 4$

15. Perform the indicated operations:$\frac{6-(-2)}{2}$.

A. 3
B. 4
C. 5
D. 6

16. Simplify: $\sqrt{[2 + (-5)] + 12}$.

A. 1
B. 3
C. 5
D. 7

17. Perform the indicated operations: $3 + \frac{\frac{4-2}{8}}{3-(-1)}$

A. 4
B. 8
C. 12
D. 16

18. Simplify: $\frac{5-(-4)+6}{\sqrt{2-(-4)+19}}$

A. 9
B. 6
C. 3
D. 0

19. Simplify: $12 + 23 - [4(4-2)]$

A. 32
B. 22
C. 12
D. 27

20. Simplify: $23 + 34 \div 3 + 34 \div 3 - 52$.

A. 6.3
B. -6.3
C. 3.6
D. -3.6

2. Solve problems involving
a. GCF or LCM
b. prime and composite numbers
c. divisibility
d. inverse or partitive proportions
e. congruence
f. linear Diophantine equation

21. If $d = \gcd(a, b)$, then the following numbers are divisible by d except for:

A. $a + b$
B. $a \times b$
C. $a - b$
D. $a \div b$

22. Compute $\mathrm{lcm}(\gcd(45,18), 5)$.

A. 45
B. 18
C. 9
D. 3

23. Which of the following pairs of number are relatively prime?

A. 27 and 48
B. 9 and 51
C. 27 and 66
D. 31 and 53

24. If $\gcd(a, b) = 1$, which of the following statements is true about $\mathrm{lcm}(a, b)$?

A. $\mathrm{lcm}(a, b) = ab$
B. $\mathrm{lcm}(a, b) = \min\{a, b\}$
C. $\mathrm{lcm}(a, b) = \max\{a, b\}$
D. $\mathrm{lcm}(a, b) = 1$

25. Find the positive factors of 90 such that one factor is 3 less than a number and the other factor 6 more than the same number.
    A. 9 and 10
    B. 18 and 5
    C. 6 and 15
    D. 2 and 45

26. Which of the following composite numbers is a power of exactly one prime number?
    A. 343
    B. 100
    C. 432
    D. 250

27. Which of the following statements is **INCORRECT**?
    I. All prime numbers are odd.
    II. 1 is a prime number.
    III. All even numbers are not prime.

    A. I only
    B. II only
    C. III only
    D. I, II, and III.

28. For an integer n, $2^{4n} + 3$ is divisible by 4 when
    I. n is negative.
    II. n is positive.
    III. n is 0.

    A. I only
    B. II only
    C. III only
    D. I, II, and III.

29. Which of the following numbers is divisible by 11?
    A. 1024
    B. 1551
    C. 1768
    D. 1912

30. Suppose that a number is divisible by 3, which of the following statement is **CORRECT**?
    I. If it is divisible by 6 then it is not divisible by 9.
    II. If it is an odd number then it is divisible by 9.
    III. If it is an even number then it is divisible by 6.
    A. I only
    B. II only
    C. III only
    D. I, II, and III

31. $1 + 2 + \cdots + 99 + 100$ is always divisible by the following odd numbers **EXCEPT**
    A. 3
    B. 5
    C. 11
    D. 17

32. Which of the following congruence is **NOT TRUE?**
A. $5 \equiv 252 \bmod(13)$
B. $15 \equiv -42 \bmod(19)$
C. $7 \equiv 73 \bmod(11)$
D. $10 \equiv -62 \bmod(11)$

33. How many distinct incongruent solutions are there in $98x \equiv 175 \bmod(49)$?
A. 8
B. 7
C. 6
D. 5

34. Determine the last digit in 8^{100}.
A. 2
B. 4
C. 6
D. 8

35. Which of the following has an **NO** integral solution?
A. $36x \equiv 432 \bmod(42)$
B. $28x \equiv 13 \bmod(75)$
C. $32x \equiv 52 \bmod(44)$
D. $21x \equiv 49 \bmod(27)$

36. How many distinct incongruent solutions are there in $-24x \equiv 52 \bmod(76)$?
A. 4
B. 5
C. 6
D. 7

37. What is the last digit if the number $17^{16} + 2$ is expanded?
A. 9
B. 7
C. 5
D. 3

38. What is the remainder when $1! + 2! + 3! + \cdots + 100!$ is divided by 8?
A. 1
B. 3
C. 4
D. 6

39. What is the particular solution of $30x \equiv 18 \bmod(72)$?
A. 7
B. 5
C. 3
D. 1

40. Find the solution of the following linear congruences.

$$x \equiv 7 \bmod(9)$$
$$x \equiv 8 \bmod(10)$$
$$x \equiv 9 \bmod(11)$$

A. $x = 988$
B. $x = 278$
C. $x = 128$
D. $x = 458$

41. What is the solution of the linear congruences below?

$$4x \equiv 5 \bmod(13)$$
$$x \equiv 3 \bmod(9)$$
$$7x \equiv 6 \bmod(8)$$

A. $x = 462$
B. $x = 570$
C. $x = 215$
D. $x = 665$

42. Find the largest value of x such that $x^2 \equiv -5 \bmod(9)$.

A. $x = 7$
B. $x = 6$
C. $x = 8$
D. $x = 5$

43. Which of the following has **NO** integral solution?

A. $13x + 17y = 24$
B. $12x + 4y = 7$
C. $15x + 24y = 99$
D. $x + y = 1$

44. Which of the following is the particular solution for $7x + 4y = 62$?

A. $x = -56, y = 110$
B. $x = -40, y = 82$
C. $x = -62, y = 124$
D. $x = -48, y = 96$

45. A restaurant offered an eat-all-you-can menu last January. They charged 250 pesos for kids below 3 ft. and 350 pesos for adults or kids beyond 3 ft. If the restaurant earned 12,450 on the first day, what is the minimum number of kids below 3 ft, if there are more customers who paid 350 pesos?

A. 2
B. 5
C. 7
D. 10

46. To help her family, Angelica is selling *kakanin* before she goes to school. She sells *pichi-pichi* and *bibingka* at 10.50 pesos and 12.75 pesos per slice, respectively. Her total earnings today is 389.25. If she sold more *pichi-pichi* than *bibingka*, what is the total number of *kakanin* sold?

A. 30
B. 33
C. 36
D.39

47. 1296 is divided into 4 for parts, 4:6:3:5. What are the parts?

A. 288,432,216, and 360
B. 288,504,72, and 432
C. 216, 576, 144, and 360
D. 216, 648, 144, and 288

48. Last Christmas, Tita Solita gave Nena and her two other siblings a total of 4500 pesos. Her mother decided to divide the money so that Nena's share and her two other siblings' share is $\frac{3}{4}:\frac{1}{3}:\frac{2}{5}$. How much is the share of each?
    A. 2225.2, 1036.28, and 1238.52
    B. 2075.48, 1111.44, and 1313.08
    C. 1975.10, 1161.33, and 1363.57
    D. 2275.28, 1011.24, and 1213.48

49. If 3 painters can finish the job in 7 hours, how long does it take for 2 painters to finish the same job?
    A. 5.67 hours
    B. 4.67 hours
    C. 3.67 hours
    D. 2.67 hours

50. The food storage of a military bunker is good for 200 soldiers and will last for 120 days. After 50 days, 70 men left the bunker for a mission. How long would the food last?
    A. 200 days
    B. 300 days
    C. 150 days
    D. 250 days

3. Apply Euler's function and theorems, or Fermat's theorem in solving problems

51. What is the remainder when 5^{34} is divided by 17?
    A. 6
    B. 10
    C. 8
    D. 12

52. Compute $\phi(71)$.
    A. 72
    B. 70
    C. 1
    D. 8

53. What is the remainder when 2^{36} when divided by 37?
    A. 7
    B. 33
    C. 9
    D. 0

54. Determine the remainder when $\sum_{n=1}^{50} n^3$ is divided by 7?
    A. 2
    B. 4
    C. 3
    D. 1

55. Which of the following is the remainder when 2^{52} is divided by 11?
    A. 4
    B. 9
    C. 3
    D. 10

56. Find the remainder when 5^{43} is divided by 16.
   A. 13
   B. 6
   C. 8
   D. 11

57. Which of the following statement is NOT TRUE about the $\phi(n)$?
   I. $\phi(n)$ is always even.
   II. If n is prime then $\phi(n) = n - 1$.
   III. n is always divisible by $\phi(n)$.

   A. I only
   B. II only
   C. III only
   D. I, II, and III

58. Find the remainder when $13^{210} + 3$ when divided by 15?
   A. 4
   B. 3
   C. 7
   D. 12

59. Find x in $10! \equiv x \bmod(11)$.
   A. 1!
   B. 7!
   C. 12!
   D. 9!

60. Evaluate $\phi(180)$.
   A. 90
   B. 80
   C. 68
   D. 48

61. Which of the following statement is TRUE when $\gcd(a, n) = 1$?
   I. $a^n \equiv a \bmod(n)$
   II. $a^{\phi(n)} \equiv 1 \bmod(n)$
   III. $a^{n-1} \equiv a \bmod(n)$

   A. I only
   B. II only
   C. III only
   D. I, II, and III

62. Which of the following given can be solved using Euler's Theorem?
   I. $16^8 \equiv x \bmod(20)$
   II. $7^{100} \equiv x \bmod(15)$
   III. $25^4 \equiv x \bmod(5)$

   A. I only
   B. II only
   C. III only
   D. I, II, and III

63. Which of the following is the remainder when 5^{74} is divided by 27?
A. 21
B. 23
C. 25
D. 0

64. What is the remainder when $10^{73} + 13^{42}$ is divided by 11?
A. 5
B. 3
C. 9
D. 7

65. Find the remainder in $3^{30} + 4^{40} + 5^{50} + 6^{60}$ divide by 7.
A. 6
B. 2
C. 1
D. 3

66. Suppose that $\gcd(x, 13) = 1$, solve for x in $x^{86} \equiv 3 \bmod(13)$.
A. 4 and 9
B. 5 and 8
C. 3 and 7
D. 2 and 5

67. If 1024 is divided to 11 persons such that all of them will receive an equal amount, how much is the remainder?
A. 9
B. 5
C. 3
D. 1

68. What does $\phi(n)$ count?
A. The number of integers from 1 to n which are relatively prime to n.
B. The number of integers from 1 to n which are factors of n.
C. The number of integers from 1 to n which are multiples to n.
D. The number of integers from 1 to n which are divisible by n.

69. Find x in $x \equiv 2^{15} \bmod(51)$
A. 24
B. 26
C. 28
D. 30

70. Which of the following can be a divisor of 11^{100} so that the remainder is 1?
A. 39
B. 101
C. 53
D. 42

71. Which of the following can be a value for z so that when we divide 25^z by 36, the remainder is 1?
A. $z = 13$
B. $z = 36$
C. $z = 37$
D. $z = 14$

72. What is x in $9^{16} \equiv x \bmod(17)$?
A. 12
B. 8
C. 5
D. 1

73. Find the remainder when 8^{22} is divided by 3?
A. 0
B. 1
C. 2
D. 3

74. What is the remainder of 13^{58} when divided by 99?
A. 13
B. 58
C. 1
D. 99

75. Find the remainder in $(8^{56} - 3) \div 9$.
A. 1
B. 3
C. 5
D. 7

Basic and Advanced Algebra

1. Perform operations on algebraic expressions

76. Which of the following is the product of $(2x^{2a}+3x^{5a})$ and $(3x^{7a}-5x^{10a})$?

A. $15x^{15a}-x^{12a}+6x^{9a}$

B. $15x^{15a}+x^{12a}+6x^{9a}$

C. $-15x^{15a}-x^{12a}+6x^{9a}$

D. $-15x^{15a}-x^{24a}+6x^{9a}$

77. What is the sum of $(x^5+2x^3+3x^2+7)$ and $(4x^4-9x^2-x-15)$?

A. $x^5+2x^3-12x^2-x-8$

B. $x^5+4x^4+2x^3-6x^2-x-8$

C. $x^5+4x^4+6x^2-x-8$

D. $x^5+6x^4-6x^2-x-8$

78. What is the quotient when $\left(\dfrac{36a^3b^5}{-24bc^3}\right)$ is divided by $\left(\dfrac{15a^2b^3}{-20ac^2}\right)$?

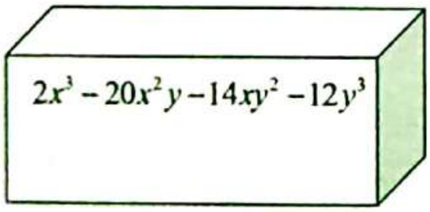
A. $-\dfrac{2a^2b}{c}$

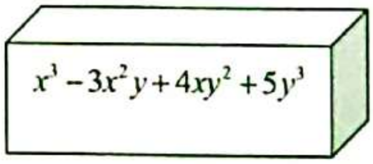
B. $\dfrac{2ab}{c^2}$

C. $-\dfrac{c}{2a^2b}$

D. $\dfrac{2a^2b}{c}$

79. Find the difference between the volumes of the two polyhedrons below.





80. If the cost of a basket of apples $(21x^3 - 18x^2 - 120x)$ and there are $(3x - 6)$ apples, how much is a piece of apple?

A. $7x + 20$
B. $-7x^2 + 20x$
C. $7x^2 - 20x$
D. $7x^2 + 20x$

2. Simplify a given expression with series of operations

81. What is the simplest form of the expression $[4(x-3)]-[2(y-3)-5(y+8)]$?

A. $4x + 3y - 34$
B. $-4x - 3y + 34$
C. $-4x + 3y - 34$
D. $4x + 3y + 34$

82. Which of the following is the simplest form of the expression $5+2^3+3[6-3(4-2)]$?

A. 13
B. 8
C. 5
D. 0

83. What is the simplest form of $\left\{m\left[m(m)^{\frac{2}{3}}\right]^{\frac{2}{3}}\right\}^{\frac{2}{3}}$?

A. $m^{\frac{19}{9}}$
B. $m^{\frac{38}{27}}$
C. $m^{\frac{47}{27}}$
D. $m^{\frac{20}{27}}$

84. If x = 2 and y = -1, what is the value of the expression $x^3 \div 4xy + 2x + 4y \div y^2$?

A. 0
B. – 1
C. 4
D. 1

85. What is the value of the expression $\frac{(x^2+y^2-3x+4y)^2}{3}+\frac{(2xy)^2}{3}$, if x = -2 and y = 1?

A. $\frac{-241}{3}$
B. $\frac{11}{3}$
C. $\frac{241}{3}$
D. $\frac{-11}{3}$

3. Simplify rational expressions using a) Laws of exponents and b) Factoring

86. What is the simplest form of the rational expression $\frac{2^{-3}m^{-5}n^{-2}}{4^{-2}m^{-4}n^{-1}}$?

A. $\frac{2m}{n}$ B. $2mn$ C. $\frac{mn}{2}$ D. $\frac{2}{mn}$

87. Which is the simplest form of $\left(\frac{1}{2}\right)^{-a}\cdot\left(8^{-b}\right)$?

A. 4^{ab} B. 2^{3ab} C. 4^{3a-b} D. 2^{a-3b}

88. Which is the simplest form of $\frac{a^{-2}+b^{-2}}{(ab)^{-2}}$?

A. $a+b$ B. a^2b^2 C. a^2+b^2 D. $\frac{b+a}{ab}$

89. Which of the following is equivalent to $\frac{-x+3}{4-x}$?

A. $\frac{3}{4}$

B. $\frac{x-3}{x-4}$

C. $\frac{x-3}{-x+4}$

D. $-\left[\frac{-(x-3)}{-(x+4)}\right]$

90. What is the simplest form of $\left(-\frac{1}{27}\right)^{-\frac{2}{3}}+\left(-\frac{1}{32}\right)^{-\frac{2}{5}}$?

A. – 13 B. – 5 C. 5 D. 13

91. Which of the following is the simplified expression of $\frac{5x^2y^5}{25xy}$?

A. $\frac{xy^4}{5}$ B. $\frac{xy}{5}$ C. $\frac{5}{xy^4}$ D. $5xy^4$

92. Simplify the expression $\frac{x+3}{x^2+12x+27}$?

A. $x+9$ B. $\frac{1}{x+9}$ C. $\frac{x+9}{x+3}$ D. $\frac{x+3}{x+9}$

93. The area of a rectangle is x^2+5x+6. The width of a rectangle is $x+2$. Determine the ratio of the area to the width.

A. $(x+2)(x+3)$ B. $x+2$ C. $x+3$ D. $\frac{1}{x+3}$

94. An isosceles triangle has two sides of length $3x+5$. The perimeter of the triangle is $9x+15$. Determine the ratio of the base to the perimeter.

A. $\frac{1}{3}x$ B. $\frac{1}{3}$ C. 3 D. $3x$

95. The width of a rectangle is $6x+8$ and the length of the rectangle is $12x+16$. Determine the ratio of the width to the perimeter.

A. $\frac{1}{3}$ B. 3 C. $\frac{1}{6}$ D. 6

4. Identify the domain and/or the range of a given function

96. Which of the following sets of ordered pairs defines relation as a function?

A. $\{(-3,1),(3,3),(4,3),(1,3)\}$

B. $\{(3,1),(-3,3),(3,3)\}$

C. $\{(-3,1),(-3,3),(6,3),(-1,3)\}$

D. $\{(\pi,1),(\pi,3),(\pi,3)\}$

97. Which of the following is the domain of the function $f(x)=\frac{3x-2}{3x-5}$?

A. $\{x \in R \mid x \neq \frac{5}{3}\}$ B. $\{x \in R \mid x \neq 0\}$ C. $\{x \in R \mid x \neq -\frac{5}{3}\}$ D. $\{x \in R \mid x \neq 5\}$

98. Let $a, b \in R$, which of the following does not define relation as function?

A. $\{(a,b) \mid a = b\}$

B. $\{(a,b) \mid a = |b|\}$

C. $\{(a,b) \mid b = (a-1)^2\}$

D. $\{(a,b) \mid a = 5b + 2\}$

99. What is the range of the function $f(x)=3x-5$ given the domain is the set of whole numbers less than 4?

A. $\{-5,-2,1,4\}$

B. $\{1,4,7,10\}$

C. $\{-8,-5,-2,1\}$

D. $\{-2,1,4,7\}$

100. The domain and range, respectively, of the function $f(x)=\sqrt{x+4}-6$ is ________.

A. $\{x \in R \mid x \neq -4\}$ and $\{y \in R \mid y \geq -6\}$

B. $\{x \in R \mid x \neq -4\}$ and $\{y \in R \mid y \neq -6\}$

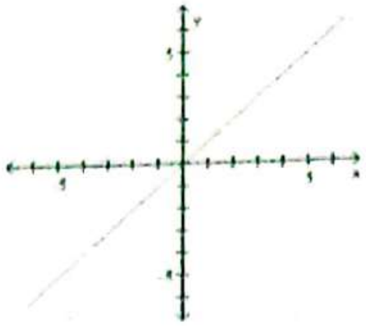
C. $\{x \in R \mid x \geq -4\}$ and $\{y \in R \mid y \leq -6\}$

D. $\{x \in R \mid x \geq -4\}$ and $\{y \in R \mid y \geq -6\}$

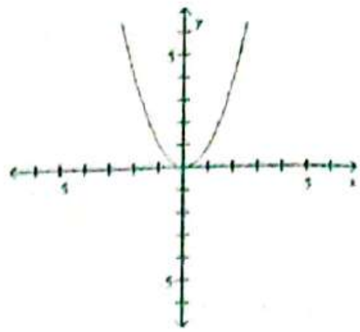
5. Identify/describe graphs of function

For numbers 101-103. Refer to the graphs below.

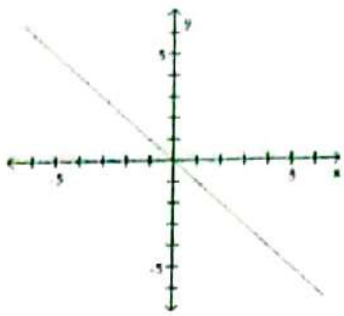
A.



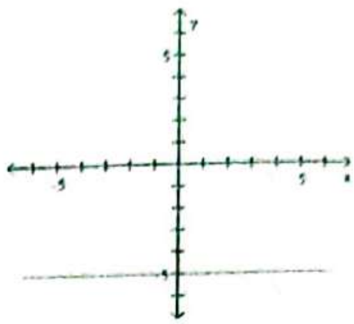
C.



B.



D.



101. Which graph shows $f(x) = x$?

102. Which graph represents a linear function with zero slope?

103. Which graph shows a quadratic equation?

104. Which of the following has a graph that opens downward?

A. $f(x)=7x^4$ C. $f(x)=-y^2$
B. $f(x)=y^2$ D. $f(x)=-7x^4$

105. Which of the following has a graph that opens upward?
A. $f(x)=7x^3$ C. $f(x)=-y^2$
B. $f(x)=x^2$ D. $f(x)=-7x^4$

6. Solve for the roots of a given quadratic equations

106. What are the roots of $4x = x^2 - 45$?

A. $\{5,-9\}$ B. $\{5,9\}$ C. $\{-5,9\}$ D. $\{-5,-9\}$

107. For what value of k will the equation $x^2 - 8x + k$ have the equal roots?

A. 2 B. 16 C. 8 D. 4

108. What is the quadratic equation whose roots are $\frac{1}{2}$ and $\frac{3}{4}$?

A. $3x^2 - 6x - 15 = 0$

B. $x^2 + 8x - 15 = 0$

C. $8x^2 - 10x + 3 = 0$

D. $4x^2 - 4x - 5 = 0$

109. What are the roots of the quadratic equation $12x^2 - x = 6$?

A. $\left\{-\frac{2}{3}, \frac{3}{4}\right\}$

B. $\left\{\frac{2}{3}, \frac{3}{4}\right\}$

C. $\left\{\frac{2}{3}, -\frac{3}{4}\right\}$

D. $\left\{-\frac{2}{3}, -\frac{3}{4}\right\}$

110. What is the quadratic equation whose roots are -3 and 6?

A. $x^2 - 3x = -18$

B. $x^2 + 3x = 18$

C. $x^2 + 3x = -18$

D. $x^2 - 3x = 18$

7. Use factoring in simplifying rational expressions

111. What is $\left(\frac{m^2-4n^2}{2m+4n}\right)\cdot\left(\frac{8m-4n}{m^2-mn-2n^2}\right)\cdot\left(\frac{m^2-4mn+3n^2}{2m^2-7mn+3n^2}\right)$ in simplest form?

A. $\frac{2m+2n}{m-n}$

B. $\frac{m-n}{m+n}$

C. $\frac{2m-2n}{m+n}$

D. $\frac{m+n}{2m-2n}$

112. An isosceles triangle has two sides of length $4x+6$. The perimeter of the triangle is $14x+21$. Determine the ratio of the base to the perimeter.

A. $\frac{3}{7}$ B. $\frac{7}{3}$ C. $\frac{3}{7}x$ D. $\frac{7}{3}x$

113. Ivan and Javier manipulated the rational expression $\frac{6x+4}{12x+4}$. Their responses are shown below.

Ivan	Javier
$\frac{3x+2}{6x+2}$	$\frac{6x}{12x}$

Which student wrote an expression that is equivalent to the original expression?

A. Only Ivan

B. Only Javier

C. Both Ivan and Javier

D. Neither Ivan nor Javier

114. Which of the following is the simplified expression of $\frac{x^2+8xk+16k^2}{x^2-16k^2}$?

A. $\frac{x-4k}{x+4k}$ B. 1 C. $x+4k$ D. $\frac{x+4k}{x-4k}$

115. Which of the following is the simplified expression of $\frac{3x^2-9xy-12y^2}{6x^3-6xy^2}$?

A. $\frac{x-4y}{2x-y}$ B. $\frac{x-4y}{2x}$ C. $\frac{2x(x-y)}{x-4y}$ D. $\frac{x-4y}{2x(x-y)}$

8. Perform operations on rational expressions

116. Simplify $\frac{x+4}{x^2+6x+8}$.

A. $\frac{1}{x+2}$ B. $\frac{1}{x+4}$ C. $\frac{1}{x+6}$ D. $\frac{1}{x+8}$

117. Simplify $\frac{8-j}{j^2-64} \div \frac{j-8}{j+8}$.

A. $\frac{1}{j-8}$ B. $\frac{-1}{j-8}$ C. $\frac{1}{j+8}$ D. $\frac{-1}{j+8}$

118. Simply $\left(\frac{3p^2+7p+2}{p^2-2p-8}\right)\left(\frac{p^2-3p-4}{p^2+3p+2}\right)$.

A. $\frac{3p+1}{p+2}$ B. $\frac{3p-1}{p+2}$ C. $\frac{3p+1}{p-2}$ D. $\frac{3p-1}{p-2}$

119. Simplify $\left(\frac{5z^2+z}{z^2-1}\right) \div \left(\frac{10z^2+2z}{4z^2+z-5}\right)$.

A. $\frac{4z-5}{2z+2}$ B. $\frac{4z+5}{2z-2}$ C. $\frac{4z+5}{2z+2}$ D. $\frac{4z-5}{2z-2}$

120. Simply $\frac{7}{x^2-64} + \frac{3}{x+8}$.

A. $\frac{3x-17}{x^2+64}$ B. $\frac{3x+17}{x^2-64}$ C. $\frac{3x+17}{x^2+64}$ D. $\frac{3x-17}{x^2-64}$

9. Perform operations on radical numbers

121. Simplify $\sqrt[4]{81a^4b^{-12}c^5}$.

A. $\frac{3ac\sqrt[4]{c}}{b^3}$ B. $\frac{3ac\sqrt[4]{c}}{b^2}$ C. $\frac{2ac\sqrt[4]{c}}{b^3}$ D. $\frac{2ac\sqrt[4]{c}}{b}$

122. Simplify $(\sqrt[3]{9x^2y})(\sqrt[3]{3x^4y^3})$.

A. $3xy\sqrt[3]{y}$ B. $x^2y\sqrt[3]{y}$ C. $3x^3y\sqrt[3]{y}$ D. $3x^2y\sqrt[3]{y}$

123. Simplify $\sqrt{72x^4y^5z^6}$.

A. $6x^2y^2z^3\sqrt{2y}$ C. $6x^2y^2z^3\sqrt{3y}$

B. $6xy^2z^3\sqrt{2y}$ D. $6x^2y^2z^2\sqrt{2y}$

124. Simplify $\frac{3-\sqrt{2}}{5+\sqrt{2}}$.

A. $\frac{15-8\sqrt{2}}{23}$ B. $\frac{17-8\sqrt{2}}{23}$ C. $\frac{17+8\sqrt{2}}{23}$ D. $\frac{15+8\sqrt{2}}{23}$

125. Simplify $\frac{3}{\sqrt[9]{x^2y^4z^7}}$.

A. $\frac{\sqrt[9]{x^7y^5z^2}}{xyz}$ B. $\frac{3\sqrt[9]{x^7y^5z^2}}{xz}$ C. $\frac{\sqrt[9]{x^7y^5z^2}}{xy}$ D. $\frac{3\sqrt[9]{x^7y^5z^2}}{xyz}$

10. Solve problems on
a) linear equations
b) systems of linear equations
c) quadratic equations

126. The sum of two numbers is 25. One of the numbers exceeds the other by 9. Find the numbers.

A. 5 and 25 B. 6 and 19 C. 7 and 18 D. 8 and 17

127. The difference between two numbers is 40. The ratio of the numbers is 5:3. What are the two numbers?

A. 100 and 60 B. 90 and 50 C. 80 and 40 D. 70 and 30

128. The length of a rectangle is thrice its breadth. If the perimeter is 112 meters, find the length and breadth of the rectangle.

A. Length – 40 m
Breadth – 12 m

B. Length – 42 m
Breadth – 14 m

C. Length – 44 m
Breadth – 16 m

D. Length – 46 m
Breadth – 18 m

129. Abby is 5 years younger than Leah. Four years later, Leah will be twice as old as Abby. Find their present ages.

A. Leah – 6, Abby – 1
B. A. Leah – 8, Abby – 3
C. Leah – 10, Abby – 5
D. Leah – 12, Abby - 7

130. 105 less than 8 times a number is equal to the number. Find the number.

A. 5 B. 10 C. 15 D. 20

131. Naida has 6 less than 8 times as many skirts as Ynah. Together, Naida and Ynah have 147 skirts. How many skirts does Naida have? How many skirts does Ynah have?

A. Naida – 120 , Ynah – 27
B. Naida – 117 , Ynah – 37
C. Naida – 110 , Ynah – 77
D. Naida – 130 , Ynah – 17

132. Two planes, which are 2700 miles apart, fly toward each other. Their speeds differ by 50 miles per hour. They pass each other after 6 hours. Find their speeds.

A. One's plane speed is 205 mph. The other plane's speed is 240 mph.

B. One's plane speed is 200 mph. The other plane's speed is 250 mph.

C. One's plane speed is 195 mph. The other plane's speed is 260 mph.

D. One's plane speed is 190 mph. The other plane's speed is 270 mph.

133. Php 70,000 is divided between two accounts. One account pays 4% interest, while the other pays 3.5% interest. At the end of the period, the interest earned was Php 2500. How much was invested in each account?

A. Php 40,000 was invested at 3.5% and Php 6,000 was invested at 4%.

B. Php 50,000 was invested at 3.5% and Php 8,000 was invested at 4%.

C. Php 60,000 was invested at 3.5% and Php 10,000 was invested at 4%.

D. Php 70,000 was invested at 3.5% and Php 12,000 was invested at 4%.

134. If two equations can be drawn as simple line on graph then these equations are considered as

A. Constant equations

B. Solved equations

C. Equivalent equations

D. Non-equivalent equations

135. A set which consists of more than one equations is classified as ______.

A. system of equations

B. system of variables

C. system of constants

D. system of coefficients

136. Which system has two equations that have no values to satisfy both equations?

A. consistent system

B. inconsistent system

C. solution system

D. constant system

137. The following linear system is to be solved using a substitution method.

$$\begin{cases} x = 5y - 12 \\ 3x + 4y = 125 \end{cases}$$

After the initial substitution, the resulting expanded equation would be

A. $15y - 36 + 4y = 125$

B. $15y + 36 + 4y = 125$

C. $15y - 36 - 4y = 125$

D. $15y + 36 - 4y = 125$

138. The solution of this linear system is $(-2, y)$. Determine the value of y.

$$-x + y = 8$$

$$2x + 3y = 14$$

A. 3 B. 4 C. 5 D. 6

139. The first equation of a linear system is $12x + 5y = 125$. Choose a second equation to form a linear system with infinite solutions.

i. $12x + 5y = 125$
ii. $-48x - 20y = -500$
iii. $-48x + 20y = -500$
iv. $-48x - 20y = 126$

A. Equation iv B. Equation iii C. Equation ii D. Equation i

140. A right triangle has a side with length 16 in. and a hypotenuse with length 25 in. Find the length of the second leg. (Round to the nearest hundredth if needed).

A. 19.21 in. B. 20.21 in. C. 21.21 in. D. 22.21 in.

141. Find the values of x for the given equation.

$$x^2 - 4x - 12 = 0$$

A. $2 \text{ and } 6$ B. $-2 \text{ and } -6$ C. $-2 \text{ and } 6$ D. $2 \text{ and } -6$

142. Find the x-intercept of $y = 6x^2 - 7x - 3$.

A. $\left(-\frac{3}{2}, 0\right) \text{ and } \left(-\frac{1}{3}, 0\right)$

C. $\left(\frac{3}{2}, 0\right) \text{ and } \left(-\frac{1}{3}, 0\right)$

B. $\left(\frac{3}{2}, 0\right) \text{ and } \left(-\frac{1}{3}, 0\right)$

D. $(2, 0) \text{ and } (-3, 0)$

143. Which statement best describes the solutions to the equation below?

$$6x^2 - 10x + 40 = 0$$

A. There are 2 rational solutions

C. There are no real solutions

B. There are 2 irrational solutions

D. There is one rational solution

144. Find all the values of x for the given equation

$$\frac{1}{2}(x^2 + 5) = 20.5$$

A. ± 7 B. ± 6 C. ± 5 D. ± 4

145. The area of a rectangle is 135 in^2. Find the perimeter of the rectangle if the length of the rectangle is 3 inches less than twice its width.

A. 45 in. B. 46 in. C. 47 in. D. 48 in.

146. If the roots of a quadratic equation are -5 and 8, then the equation is

A. $x^2 - 3x - 40 = 0$

C. $x^2 - 3x + 40 = 0$

B. $x^2 + 3x - 40 = 0$

D. $x^2 + 3x + 40 = 0$

147. If the sum and product of roots of a quadratic equation are $\frac{1}{2}$ and -3 respectively, then the equation is

A. $2x^2 - x - 6 = 0$
C. $2x^2 - x + 6 = 0$
B. $2x^2 + x - 6 = 0$
D. $2x^2 + x + 6 = 0$

148. Which constant must be added or subtracted to solve the equation $x^3 + \frac{2}{3}x = 0$ by the method completing the square

A. 1/3
B. 1/5
C. 1/7
D. 1/9

149. Find three consecutive positive integers such that the sum of the square of the first and the product of the other two is 46.

A. 5 , 6 , 7
B. 4 , 5 , 6
C. 6, 7 , 8
D. 7 , 8 , 9

150. If one root of $2x^2 + kx + 6 = 0$ is $-\frac{3}{2}$, then the value of k is ______.

A. 3
B. 5
C. 7
D. 9

11. Compute the value of function f(n) where n is a counting numbers

151. What is the value of the function $f(x) = x^2 - 3x + 4$ if $x = -5$?

A. -6
C. 14
B. -5
D. 44

152. Which of the following is the value of the function $f(x) = -2x^2 + 6x - 3$ when $x = -2$?

A. 13
C. 23
B. -13
D. -23

153. The value of $h(x) = x^2 - 4x + 9$ when $x = 3$ is ___________.

A. 3
B. 6
C. 9
D. 12

For numbers 154-155: $y = 3x + 2$

x	1	2	3	4
y	5			14

154. What is the value of y when *x=2*?

A. 8
B. 9
C. 12
D. 15

155. What is the value of y when *x=3*?

A. 7
B. 9
C. 11
D. 13

12. Determine an equation given a set of which are imaginary/complex numbers

156. Which of the following is an imaginary number?

A. 3.14
B. $\sqrt{5}$
C. $\sqrt{-5}$
D. π

157. What are the roots of the equation $x^2 + 4x + 5 = 0$?

A. $-1 \pm i$
B. $-2 \pm i$
C. $1 \pm i$
D. $2 \pm i$

158. Find the solutions of the quadratic equation $14x^2+9x+10=0$.

A. $\frac{9}{28} \pm \frac{\sqrt{479}}{28}i$
B. $-\frac{9}{28} \pm \frac{\sqrt{479}}{28}i$
C. $\frac{9}{18} \pm \frac{\sqrt{479}}{18}i$
D. $-\frac{9}{18} \pm \frac{\sqrt{479}}{18}i$

159. Find the solutions of the quadratic equation $x^2 - 5x - 10=0$.

A. $\frac{5}{12} \pm \frac{\sqrt{65}}{12}i$
B. $-\frac{5}{12} \pm \frac{\sqrt{65}}{12}i$
C. $\frac{5}{2} \pm \frac{\sqrt{15}}{2}i$
D. $-\frac{5}{2} \pm \frac{\sqrt{15}}{2}i$

160. Find the solutions of the quadratic equation $7x^2 - 2x + 3 = 0$.

A. $-\frac{2}{7} \pm \frac{2\sqrt{5}}{7}i$
B. $-\frac{1}{7} \pm \frac{2\sqrt{5}}{7}i$
C. $-\frac{2}{7} \pm \frac{1\sqrt{5}}{7}i$
D. $-\frac{1}{7} \pm \frac{1\sqrt{5}}{7}i$

13. Perform operations involving exponential and logarithmic functions

161. Expand: $\log\frac{x}{y}$.

A. $\frac{\log x}{\log y}$
B. $\frac{\log y}{\log x}$
C. $\log y + \log x$
D. $\log x - \log y$

162. Condense: $\log 3x + \log 2x - \log 6$.

A. $\log x^2$
B. $\log x$
C. $\log 2x$
D. $\log 2$

163. Solve for x in the equation $2^{4x} = 32$.

A. $\frac{3}{4}$
B. $\frac{5}{4}$
C. $\frac{1}{2}$
D. $\frac{3}{2}$

164. Evaluate: $\log 100$.

A. 1
B. 2
C. 3
D. 4

165. What is the value of x in the equation $3^{2x} \cdot 9 = 1$?

A. 1
B. 0
C. – 1
D. – 2

14. Solve for the solution set of a given inequality

166. Solve: $\frac{x+2}{x-1} > 0$.

A. $(-\infty, -2)$
B. $(-\infty, -2) \cup (1, +\infty)$
C. $(5, +\infty)$
D. $(-\infty, 0) \cup (4, +\infty)$

167. Solve: $\frac{x+5}{x-3} \leq 0$.

A. $(-\infty, -5] \cup (3, +\infty)$
B. $(-\infty, -5) \cup [3, +\infty)$
C. $(-5, 3]$
D. $[-5, 3)$

168. Solve: $\frac{3}{x-2} + \frac{1}{x} > 0$.

A. $(-\infty, \frac{1}{2})$
B. $(0, \frac{1}{2})$
C. $(0, \frac{1}{2}) \cup (2, +\infty)$
D. $\left(-\infty, \frac{1}{2}\right] \cup [2, +\infty)$

169. Solve: $3x - 4 \geq 0$.

A. $x > 4$
B. $x \leq 3$
C. $x \leq \frac{3}{4}$
D. $x \geq \frac{4}{3}$

170. Solve: $\frac{3}{x} < 2$.

A. $(-\infty, 0)$
B. $(0, \frac{3}{2})$
C. $(0, \frac{1}{2}) \cup (\frac{3}{2}, +\infty)$
D. $(-\infty, 0) \cup (\frac{3}{2}, +\infty)$

15. Determine the rth term of the expansion $(a \pm b)^n$

171. What is the last term in the expansion of $(2x + y)^3$?

A. $3y$
B. y^2
C. $2y^3$
D. y^3

172. What is the 4^{th} term in the binomial expansion $(3x + 1)^5$?

A. $90x^2$
B. $110x^2$
C. $225x^2$
D. $300x^2$

173. What is the 7^{th} term in expansion $(x - 2y)^9$?

A. $2187x^3y^6$
B. $5376x^3y^6$
C. $7701x^6y^3$
D. $8023x^6y^3$

174. What is the last term in the binomial expansion $(m + 2n)^{10}$?

A. $996n^{10}$ C. $1267n^{10}$

B. $1024n^{10}$ D. $1500n^{10}$

175. What is the 6^{th} term in the expansion $(-y + 3x)^{12}$?

A. $67244x^2y^8$ C. $-192456x^5y^7$

B. $87299x^6y^9$ D. $-221762x^4y^{10}$

16. Solve problems involving arithmetic and geometric progression

176. What is the 8^{th} term of the geometric sequence 6, 18, 54, 162, ... ?

A. 2 C. 174

B. 32 D. 39366

177. Find a_1 for an arithmetic sequence in which a_{10}=2 and d=-1.

A. -16 C. 7

B. -7 D. 11

178. What is the sum of the first five terms of the geometric progression in which a_1= -2 and $r = -\frac{1}{2}$?

A. $-\frac{10}{3}$ C. 1

B. $-\frac{2}{81}$ D. $-\frac{11}{8}$

179. Find the sum of an arithmetic series in which a_1=5, a_4=7 and n=9.

A. 189 C. 131

B. 172 D. 130

180. Find a_7 in an arithmetic sequence for which $a_4 = -2$ and $a_9 = 13$.

A. 4 C. 6

B. 5 D. 7

181. The three arithmetic means between 4 and 32 are ________.
A. 18, 25, 28
B. 11, 18, 25
C. 10, 18, 26
D. 10, 16, 22

182. What is the sum an arithmetic progression whose first and last term are 7 and 46, respectively, and whose 8th term is 28?
A. 14
B. 371
C. 424
D. $\frac{689}{2}$

183. What is the first term of a geometric progression when the sum of its 5 terms is equal to $\frac{31}{4}$ and the common ratio is $\frac{1}{2}$?
A. 1
B. 2
C. 3
D. 4

184. Barangay Sipnayan has a current population of 50000 and the population is increasing by 3% each year. What will be its population (approximate) in 10 years?
A. 51 546
B. 54 165
C. 56 145
D. 51 645

185. Ian Gabriel saved Php50.00 in January. Suppose he saved twice the amount he saved the previous month. How much would he save at the end of the year?
A. Php 4 046
B. Php 51 200
C. Php 102 400
D. Php 204 750

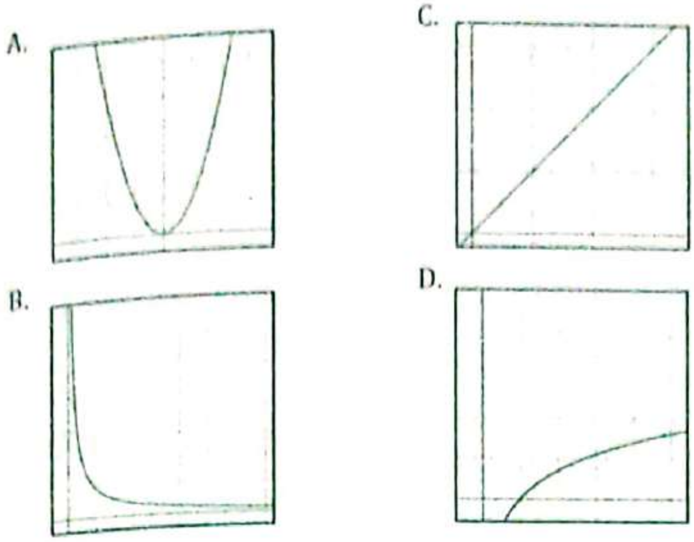
17. Solve problems involving variations

For the numbers 186 to 190, consider the table below:

No. of hours travelled (t)	1	2	3	4	5
Distance in km covered (y)	50	100	150	200	250

186. Which of the following equations represents the table above?
A. $y = 50t$
B. $y = 100t$
C. $y = 150t$
D. $y = 200t$

187. Which of the following graphs represents the table above?



188. Find y when $t = 3.5hrs$.

A. 125 km
B. 175 km
C. 215 km
D. 225 km

189. Find y when $t = 4.5hrs$.

A. 125 km
B. 175 km
C. 215 km
D. 225 km

190. Find y when $t = 10hrs$.

A. 250 km
B. 500 km
C. 750 km
D. 1 000 km

For numbers 191 to 193, find an equation of variation for the given situation:

191. m varies directly as the square of n, and m = 150 when n = 5 .

A. $m = \frac{1}{6}n^2$
B. $m = 6n^2$
C. $m = 30n^2$
D. $m = \frac{1}{30}n^2$

192. w varies inversely as the square of z, and w = 6 when z = 2.

A. $w = 4z^2$
B. $w = \frac{4}{z^2}$
C. $w = 24z$
D. $w = \frac{24}{z^2}$

193. a varies inversely as the square root of b, and a = 10 when b = 25.

A. $a = \frac{50}{\sqrt{b}}$ C. $a = \frac{50}{b}$

B. $a = \frac{100}{\sqrt{b}}$ D. $a = \frac{100}{b}$

194. The pitch (x) of a musical tone varies inversely as its wavelength (y). One tone has a pitch of 268 vibrations per second and a wavelength of 8 ft. Find the wavelength of another tone that has a pitch of 300 vibrations per second.

A. 4.25 ft C. 7.15 ft

B. 6.17 ft D. 9.30 ft

195. The distance (a) that a spring is stretched by a hanging object varies directly as the weight (b) of the object. If a 15-g object stretches a spring 8 m, how far will a 40-g weight stretch the spring?

A. 11.15 m C. 31.80 m

B. 21.32 m D. 41.74 m

18. Determine the number of positive and negative roots of a given polynomial

196. Determine the correctness of these sets of statements:

I. The number of positive real zeros of P(x) with the terms arranged in decreasing powers of x is equal to the number of variations in sign of P(x) or is less than this number by a positive even integer.

II. The number of negative real zeros of P(x) with the terms arranged in decreasing powers of x is equal to the number of variations in sign of P(-x) or is equal to that number decreased by an odd integer.

A. Only statement I is true. C. Both statements I and II are true.

B. Only statement II is true. D. Neither statement I nor II is true.

197. Using Descartes' Rule of Sign, what is the number of positive roots for the polynomial $P(x) = 6x^4 + 23x^3 - 20x^2 - 8x - 4$?

A. 0 C. 2

B. 1 D. 3

199. Determine the number of negative roots of the polynomial $P(x) = x^3 + 7x^2 + 14x + 8$.

A. 0
B. 1
C. 2
D. 3

200. For the polynomial $P(x) = x^3 + 7x^2 + 14x + 8$, how many positive real roots are there?

A. 0
B. 1
C. 2
D. 3

Plane and Solid Geometry

1. Apply the properties of plane figures such as segments, angle triangles, quadrilaterals and other polygons in solving problems

201. If two sides of an equilateral triangle measure 20 centimeters, the third side is how many centimeters long?

A. 15 B. 10 C. 5 D. 20

202. In figure 1, the value of x is ______.

A. 50
B. 60
C. 65
D. 55

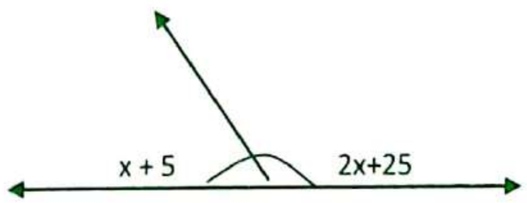


Figure 1

203. Located along a straight road, Town X to Town Z is 53km. If Town Y is midway between the first two towns, how far away is Town Y from Town Z and why?
A. 53 km, the distances are equal.
B. 53 km, the distances are the same.
C. 26.5 km, by definition of congruence.
D. 26.5 km, by the definition of midpoint.

204. A plane is determined by ________________.
I. any three non collinear points.
II. a line and a point.
III. two intersecting lines.
IV. a line and a point not on it.
A. III only B. IV only C. I and III D. III and IV

205. Find the measure of an angle if the measure of its supplement is 39 more than twice the measure of its complement.
A. 39 B. 38 C. 60 D. 120

206. The sum of the measures of an acute angle and an obtuse angle is 140^0. The sum of twice the supplement of the obtuse angle and three times the complement of the acute angle is 340^0. Find the measures of the angles.
A. $120^0, 20^0$ C. $10^0, 40^0$
B. $130^0, 10^0$ D. $110^0, 30^0$

207. If $\triangle ABC$ is equiangular, then it is ________________.
A. right B. scalene C. obtuse D. equilateral

208. Given an equilateral $\triangle$ XYZ with midpoints A,B,C respectively, what figure is formed by quadrilateral XABZ?
A. rectangle B. trapezoid C. rhombus D. square

209. In $\triangle ABC$, D and E are points on $\overline{AC}$ and $\overline{BC}$, respectively, such that $\overline{DE} // \overline{AB}$. If AC = 11, DC = 6 and BC = 15, what is the length of $\overline{BE}$?

A. 15 B. 11 C. 21.21 D. 6.82

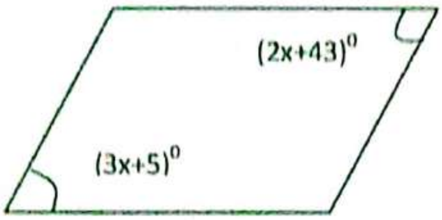
210. Which of the following BEST represents a line segment?
A. Dot
B. Edge of a table
C. Page of a notebook
D. Intersection of two planes

211. One of the angles of a parallelogram is 127.5. What are the measures of the remaining angles of the parallelogram?
A. 127.5, 127.5, 127.5, 127.5
B. 52.5, 52.5, 37.5, 37.5
C. 127.5, 52.5, 127.5, 52.5
D. 127, 50, 127

212. Which of the following is **TRUE**?
A. Every rhombus is a square.
B. Every rectangle is a square.
C. Every rectangle is a trapezoid.
D. Every square is a parallelogram

213. In an isosceles triangle, the vertex angle is half the measure of a base angle. What is the measure of the base angle?
A. 30 B. 60 C. 36 D. 72

214. In the joining figure, the quadrilateral is a parallelogram. What is the value of x?
A. 38 B. 48 C. 7.6 D. 9.6



215. Given $\triangle$ MNO with midpoints X, Y, Z, respectively, what figure is formed by quadrilateral XNOZ?
A. equilateral $\triangle$
B. Isosceles triangle
C. Isosceles trapezoid
D. parallelogram

216. Which of the following statement does **NOT** belong to the group?
  A. The diagonals of a trapezoid are congruent.
  B. The diagonals of a parallelogram are congruent.
  C. The diagonals of a square are perpendicular to each other.
  D. The diagonals of a rectangle are perpendicular to each other.

217. What is the greatest number of lines that may be drawn through three noncollinear points?
  A. 3 B. 2 C. 1 D. Infinite

218. Which of the following is the sum of the angles of a hexagon?
  A. 900^0 B. 720^0 C. $1,080^0$ D. 360^0

219. If two sides of an equilateral triangle measure 10 cm, the third side is how many cm long?
  A. 20 B. 15 C. 10 D. 5

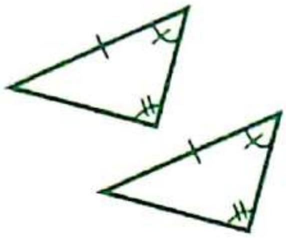
220. Which two quadrilaterals have congruent consecutive angles?
  A. Rectangle and Rhombus
  B. Parallelogram and Square
  C. Parallelogram and Rectangle
  D. Square and Rectangle

2. Use properties of congruence in justifying propositions on triangle congruence

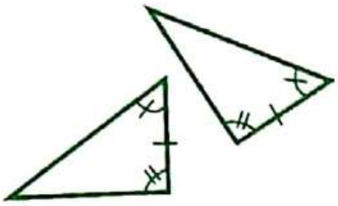
221. Which of the following postulates states that, if two sides and the included angle of one triangle are congruent to two sides and the included angle of another triangle, then the two triangles are congruent?
  A. ASA Postulate
  B. SSS Postulate
  C. SAS Postulate
  D. AAS Postulate

222. Which of the following postulates states that, if three sides of one triangle are congruent to three sides of another triangle, then the two triangles are congruent?
  A. ASA Postulate
  B. SSS Postulate
  C. SAS Postulate
  D. AAS Postulate

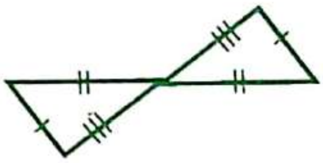
223. Which postulate of triangle congruence is present in the given triangles to the right?
A. ASA Postulate
B. SSS Postulate
C. SAS Postulate
D. AAS Postulate

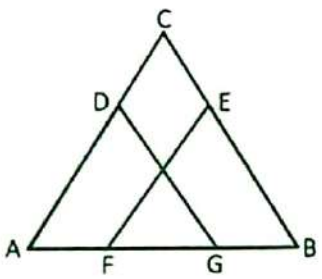


224. Which postulate of triangle congruence is present in the given triangles to the right?
A. ASA Postulate
B. SSS Postulate
C. SAS Postulate
D. AAS Postulate



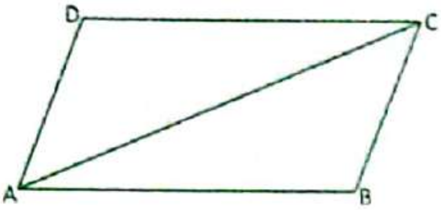
225. Which postulate of triangle congruence is present in thegiven triangles to the right?
A. ASA Postulate
B. SSS Postulate
C. SAS Postulate
D. AAS Postulate





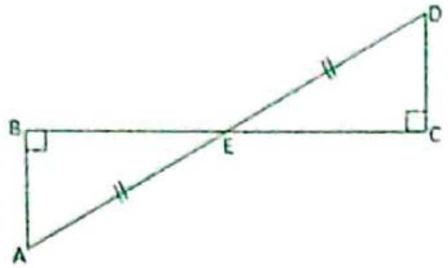
226. In the adjoining figure AF = BG, $\angle A \cong \angle B$ and AD = BE,. to prove that $\Delta DAG \cong \Delta EBF$, what congruence postulate should be applied?

A. SSS
B. SAS
C. ASA
D. SAA



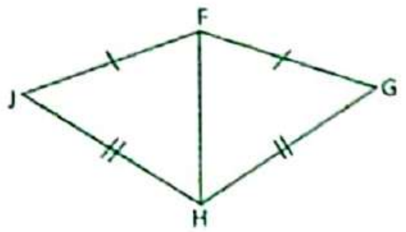
227. In the adjoining figure, $\overline{AB}$ □ $\overline{CD}$, what congruence principle may be used to show that $\Delta ADC \cong \Delta CBA$?

A. SAS
B. ASA
C. SSS
D. SAA



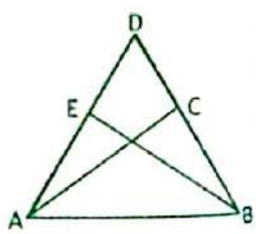
228. In the adjoining figure, $\overline{CE} = \overline{BE}$, what congruence principle may be used to show that $\Delta ABE \cong \Delta CDE$?

A. SSS
B. SAS
C. ASA
D. SAA



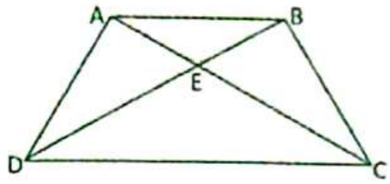
229. What congruence principle can explain why $\Delta FGH \cong \Delta FJH$?

A. ASA
B. SAA
C. SAS
D. SSS



230. In the given figure, what congruence postulate can prove that $\Delta DAC \cong \Delta DEB$ given that $\overline{AC} = \overline{BE}$, $\angle C \cong \angle E$ and $\angle A \cong \angle B$?

A. SAA
B. ASA
C. SAS
D. SSS

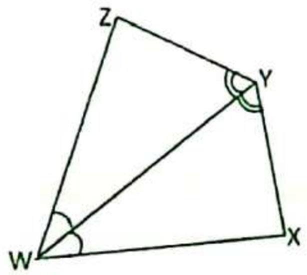


231. A trapezoid ABCD is given where $\Delta DAB \cong \Delta CBA$. Which of the following statements is TRUE based on the given information?

A. $\overline{BD} \cong \overline{CD}$
B. $\overline{DA} \cong \overline{DC}$
C. $\angle BCE \cong \angle DAB$
D. $\angle ACB \cong \angle BDA$

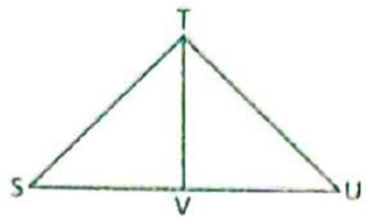
232. To prove that $\Delta QRS \cong \Delta LMN$ by SAS, what additional information is needed based on the given pair of diagram?

A. $\overline{NM} = \overline{BC}$
B. $\overline{LN} = \overline{AB}$
C. $\angle LNM \cong \angle QSR$
D. $\angle LMN \cong \angle QRS$



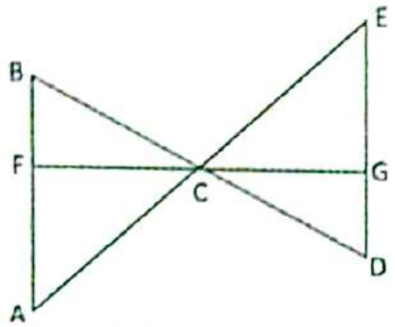
233. Which statement must be TRUE based on the given pair of congruent triangles?

A. $\angle ZWY \cong \angle WYX$
B. $\angle XYZ \cong \angle XWZ$
C. $\overline{WZ} = \overline{WX}$
D. $\overline{YZ} = \overline{WX}$



234. Which of the following statement is cannot be proven given that $\overline{TV}$ is a perpendicular bisector of $\overline{SU}$?

A. $\angle TSV \cong \angle TUV$
B. $\angle SVT \cong \angle UVT$
C. $\overline{TV} = \overline{UV}$
D. $\overline{ST} = \overline{TU}$



235. Given in the adjoining figure, $\Delta FAC \cong \Delta GEC$, which of the following statement can be proven?

A. $\angle FCA \cong \angle GCE$
B. $\angle CAF \cong \angle CDG$
C. $\overline{AF} = \overline{GE}$
D. $\overline{AC} = \overline{EC}$

3. Solve problems using definitions, postulates and/or theorems on triangles, quadrilaterals, or circles

236. From point B the angle of elevation of the top of a building at point A is 48°. From point C, 80 feet nearer the building, the angle of elevation is 75°. Find the distance of AC.

A. 309 ft.
B. 131 ft.
C. 105 ft.
D. 27 ft.

237. A 25 feet high lighthouse stands on the side of a hill. At a point of 80 feet from the foot of the lighthouse, measured straight down the hill, the lighthouse subtends an angle of $17°$. What angle does the side of the hill make with the lighthouse?

A. $44°52'$
B. $69°19'$
C. $70°8'$
D. $93°41'$

238. A surveyor found that the angle of elevation of the top of the flagpole was $61°$. The observation was made from a point 1.5 meter above the ground and 10 meter from the base of the flagpole. What is the height of the flagpole?

A. 6.35 m
B. 10.25 m
C. 18.04 m
D. 19.54 m

239. A triangular lot in front of a commercial building bounded by sides of lengths 13 meters, 11 meters, and 9 meters is to be enclosed by a fenced and planted with ornamental grass. What is the area of the lot?

A. 33.7 sq. m
B. 48.8 sq. m
C. 49.5 sq. m
D. 51.6 sq. m

240. A circle has a radius of 10 cm. If a chord of a circle is 6 cm from the center, how long is the chord?

A. 8 cm
B. 12 cm
C. 16 cm
D. 60 cm

241. Given a circle with diameter $\overline{AB}$ and cords $\overline{AC}$ and $\overline{BD}$ that intersect at E, if the measure of arc AD is 85 and the measure of arc BC is 70, what is $m\angle AEB$?

A. 155
B. 102.5
C. 90
D. 77.5

242. A farmer is decided to fence a triangular vegetable garden from his field bounded on two sides by two streets intersecting at $31°15'$. The sides of the garden along the streets are 10 meters and 5 meters. Seedlings are to be planted in the garden at a cost of Php 15.00 per sq. ft. What is the total cost?

A. Php 192.30
B. Php 319.35
C. Php 630.91
D. Php 750.00

243. An end of a trough is in the shape of an isosceles trapezoid whose area is 216 in^2. If each of the non-parallel sides is 13 inches and the height is 12 inches and one of the bases is 13 inches long, what is the length of the other base?

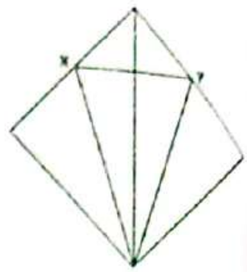
A. 23 in
B. 25 in
C. 26 in
D. 38 in

244. The ends of a piece of wood have a cross section with bases of 8 cm and 6 cm and a height of 7 cm. What is the area of the cross section?

A. 42 sq. cm
B. 48 sq. cm
C. 49 sq. cm
D. 56 sq. cm

245. The length of a piece of wire is 8 inches. It is then formed into a regular hexadecagon (polygon of 16 sides). What is the area of the polygon?
A. 2.04 sq. in
B. 5.04 sq. in
C. 7.04 sq.in
D. 10.04 sq. in

246. A kite is inscribed in a square of side 4 cm. Points X and Y are midpoints of the sides of the square. (See figure). What is the perimeter of the kite?



A. 16 cm
B. 12.94 cm
C. 8 cm
D. 6.47 cm

247. The windshield wiper of a vehicle is 32 inches long and has a silicon wiper blade of length 18 inches. During the usage of the wiper, it sweeps through 120°. What is the area swept by the blade?
A. 134.04 sq. in
B. 268.08 sq. in
C. 733.04 sq. in
D. 867.08 sq. in

248. What is the area of the shaded region in the given figure?

A. 3.43 sq. in
B. 4.00 sq. in
C. 12.57 sq. in
D. 16.00 sq. in

249. Reno is contemplating on cutting a circle, a regular octagon, or a regular decagon from a piece of cardboard whose dimensions are 12 m by 11 m. If his intention is to maximize the use of the cardboard, which among the three shapes is the best shape to produce?
A. Circle
B. Octagon
C. Decagon
D. Nonagon

250. Two sides of a parallelogram are 38 cm and 22 cm, and one of its angles is $35^{\circ}23'$. What is the length of the shorter diagonal?
A. 23.77 cm
B. 27.21 cm
C. 30.98 cm
D. 43.91 cm

251. A children's park is in the shape of a diamond (rhombus) whose diagonals are 10 ft. by 24 ft. What is the area of the park?
A. 80 sq. ft.
B. 120 sq. ft.
C. 160 sq. ft.
D. 240 sq. ft.

252. Mr. Cheng plans to carpet their family room except for a border of uniform width. The family room is 5 meters by 3 meters. What dimensions should the carpet have if it covers an area of 8 square meters?
A. 1 m by 8 m
B. 2 m by 4 m
C. 2 m by 6 m
D. 3 m by 5 m

253. Charlie has a circular garden that she separates into three equal parts. If the radius of the garden is 9 meters, what is the length of the arc of each part?

A. 2π
B. 4π
C. 6π
D. 12π

254. Darwin ran around a circular field 3 times. If he ran a total distance of 750 meters, what is the diameter of the field?

A. 159.16 m
B. 119.37 m
C. 79.58 m
D. 39.79 m

255. The diameter of a circle is 26 cm. if the angle sector is 55^0, what is the area of the sector?

A. 180^0
B. 135^0
C. 90^0
D. 45^0

4. Solve problems involving concepts on:
(a) solids, (b) similarity and proportionality, and (c) transformations.

256. A stone is dropped into a still pond. Concentric circular ripples spread out and the radius of the disturbed region increases at the rate of 16 cm/sec. At what rate does the area of the disturbed region increase when the radius is 4 cm?

A. $256\pi cm^2 / sec$
B. $128\pi cm^2 / sec$
C. $64\pi cm^2 / sec$
D. $16\pi cm^2 / sec$

257. A closed tin of volume $16\pi\ in^3$ is to be in a form of a right circular cylinder. What is the radius if the least amount of material is to be used in its manufacture?

A. 6 in
B. 4 in
C. 2 in
D. 0 in

258. A prism has a lateral edge of 3 and the perimeter of its base is 34. What is the lateral surface area of the prism?

A. 105
B. 102
C. 92
D. 37

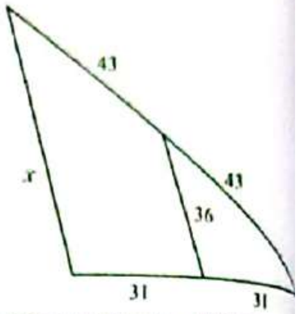
259. Given two isosceles triangles ΔABC and ΔDEF. The measure of the vertex angle of ΔABC is twice the measure of the vertex angle of ΔDEF and one-half that of a base angle of ΔDEF. What must be the measure of each of the base angles of ΔABC?

A. 40
B. 50
C. 60
D. 70

260. The vertex of a figure is located at (2, 4). The figure is rotated and the image of the vertex is located at (-2, -4). Which of the following describes the transformation of the figure?
   A. Reflection over the y-axis
   B. Reflection over the x-axis
   C. Rotation of 180 counter clockwise
   D. Rotation of 270 counter clockwise

261. A rectangular photo with dimensions of 2.5 inches wide by 5 inches long is enlarged to a length of 8 inches. What is the width of the enlarged print?
   A. 7.5 in
   B. 9.5 in
   C. 11.5 in
   D. 13.5 in

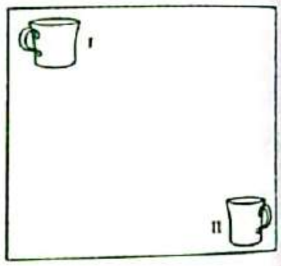
262. What is the value of x in the given diagram? Note: The diagram is not to scale.
   A. 36
   B. 62
   C. 72
   D. 86



263. The marching band enters the gym and marches across the gym without turning. Which of the following describes the transformation?
   A. Translation
   B. Reflection
   C. Rotation
   D. Dilation

264. Triangle JKL has vertices J (2, 4), K (3, 1), and L (3, 3). A translation maps the point J to J' (3, 3). What are the coordinates of K'?
   A. (4, 0)
   B. (3, 2)
   C. (2, 2)
   D. (-3, 1)

265. Jessica is a computer graphics designer and is working on an ad for the local coffee shop. The figure given shows a coffee mug in two different positions. Which of the following describes the transformation of the coffee mug in position I to the image in position II?



   A. 180° rotation
   B. Translation down and a reflection over a vertical line.
   C. A reflection over a horizontal line and a translation down.
   D. Translation to the right and a reflection over a vertical line.

266. A rectangular parallelepiped has a length of 5 cm, a width of 4 cm and a height of 8 cm. What is the total surface area of the parallelepiped?
   A. 104 sq. cm
   B. 144 sq. cm
   C. 160 sq. cm
   D. 184 sq. cm

267. A carpenter was requested by his son to construct a wooden cube that is 5 cm on a side. How much wood was used?
   A. 100 cu. M
   B. 125 cu. M
   C. 150 cu. m
   D. 175 cu. m

268. A wooden column has the form of a regular hexagonal prism. Its base edge is 1ft. and its height is 20 ft. What is the weight (in lbs.) of the column? (Hint: Use 16 lbs. per cubic ft.)

A. 480.00 lbs.
B. 678.82 lbs.
C. 831.38 lbs.
D. 960.00 lbs.

269. How many cubic yards of earth must be used to make a railroad embankment 500ft long if a cross section of the embankment is an isosceles trapezoid whose bases are 20 ft. and 45 ft. and whose legs are 18 ft. each?

A. 42,088.05
B. 64,141.35
C. 70,144.98
D. 210, 437.50

270. How many gallons of paint is/are needed to cover a room that is 4 meters long, 3 meters wide and 2.5 meters high if a gallon of paint covers about 40 sq. meters?

A. 1.475
B. 1.25
C. 0.875
D. 0.75

271. The volume of a cube is equal to the volume of a parallelepiped whose edges are 8 cm, 9 cm, and 3 cm. What is the edge of the cube?

A. 3
B. 6
C. 9
D. 12

272. A circular cylindrical tank lying on ground level has a length of 15 ft. and a diameter of 4ft. When a tank is filled to a depth of 1ft., how many cubic ft. of liquid does it contain?

A. 25.9
B. 31.4
C. 36.9
D. 62.8

273. What is the capacity of a hopper (a container that can hold and dispense bulk material such as grain, rock, or trash) in a shape of an inverted square pyramid of sides 8 ft. and 16 ft. deep?

A. 341.33 sq. ft.
B. 341.33 cu. ft.
C. 512 sq. ft.
D. 512 cu. ft.

274. A reservoir in the form of an inverted right circular cone, with diameter of 10 meters and a depth of 30 meters is being filled with water. How long will it require an inlet pipe to fill this reservoir if the water pours in at a rate of 35 cubic meters per hour?

A. 22.44 hours
B. 22.85 hours
C. 24.29 hours
D. 25.71 hours

275. A pyramidal tank has a square base of 5.5 ft. on a side and a height of 10 ft. What is the volume of water in the tank when the depth of the water is 7.35 ft.?

A. 112.45 cu. ft.
B. 113.93 cu. ft.
C. 117.62 cu. ft.
D. 120.11 cu. ft.

Trigonometry

1. Solve problems involving
a. trigonometric function values
b. trigonometric identities
c. concepts of right and oblique triangles
d. law of cosines and sine

276. Evaluate tan 240°.

C. 1 | C. $\sqrt{3}$
D. $\sqrt{2}$ | D. $\sqrt{4}$

277. What is the exact value of $\cos\ (-630)^{\circ}$?

A. 1 | C. 2
B. 0 | D. 3

278. Identify the exact value of $csc\ (-\frac{5\pi}{6})$.

C. -1 | C. 1
D. -2 | D. 2

279. Simplify $\sin\left(\frac{\pi}{4}\right)$.

A. $\frac{\sqrt{2}}{2}$ | C. $\frac{\sqrt{2}}{3}$
B. $-\frac{\sqrt{2}}{2}$ | D. $-\frac{\sqrt{2}}{3}$

280. Evaluate cos (cos 270°).

A. 0 | C. 2
B. 1 | D. $\sqrt{2}$

281. Evaluate $\tan\ (\cos(-\frac{11\pi}{2}))$.

A. 1 | C. 0
B. 2 | D. 5

282. Evaluate tan (sin300°).

A. $-\frac{\sqrt{1}}{2}$

B. $-\frac{\sqrt{3}}{2}$

C. $-\frac{\sqrt{3}}{3}$

D. $-\frac{\sqrt{3}}{5}$

283. Use the fundamental identities to simplify $\frac{\cos(\theta)\cot(\theta)}{1-\sin^2(\theta)}$.

A. $cos\ (\theta)$ B. $cot\ (\theta)$ C. $csc\ (\theta)$ D. $sin\ (\theta)$

284. Which of the following is equal to $\sec x - \cos x$?

A. $\sin x \tan x$ B. $\sin x \cot x$ C. $\sec x \tan x$ D. $\csc x \tan x$

285. Which of the following is **CORRECT?**

A. $sin^2\theta + csc^2\theta = 1$

B. $1 + tan^2\theta = sec^2\theta$

C. $1 + sec^2\theta = csc^2\theta$

D. $cot^2\theta + cot^2\theta = 1$

286. Simplify $csc^4x - cot^4 x$ using identities.

A. $csc^2x + cot^2x$

B. $\csc x + cot^2x$

C. $csc^2x + \cot x$

D. $csc^2x - cot^2x$

287. Use trigonometric identities to simplify $(\sin x + \cos x)(\tan x + \cot x)$

A. $\sin x + \csc x$ B. $\sec x + \cos x$ C. $\tan x + \csc x$ D. $\sec x + \csc x$

288. Which is a quotient identity?

A. $\sec\theta = \frac{\sin\theta}{\cos\theta}$ B. $\csc\theta = \frac{\cos\theta}{\sin\theta}$ C. $\tan\theta = \frac{\sin\theta}{\cos\theta}$ D. $\sec\theta = \frac{\tan\theta}{\cos\theta}$

289. A right triangle always has an angle of _______.

A. 90^0

B. 80^0

C. 70^0

D. 60^0

290. The total measure of the angles of an oblique triangle is _____.

A. 90^0

B. 180°

C. 270°

D. 360°

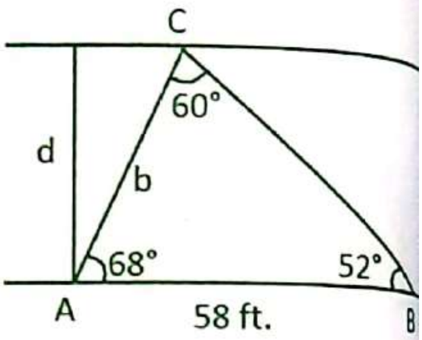
291. Which of the following triangle can never have a 90^0 angle?
A. Scalene C. Isosceles
B. Equilateral D. Oblique

292. A tree leans 7° to the vertical. At a point 40 feet from the tree (on the side closest to the lean), the angle of elevation to the top of the tree is 24°. Find the height of the tree.
A. 18.01ft C. 17.01ft
B. 20ft D. 19ft

For numbers 293 -294, refer to the problem below.

Two markers A and B, 58 feet apart, are on the same side of a river. A third marker is located across the river at point C. A surveyor determines that CAB=68° and ABC=52°.

293. What is the distance between points A and C?
A. 25.10ft C. 13ft
B. 52.78ft D. 60.5ft



294. What is the distance across the river?
A. 31.6ft C. 48.94ft
B. 43.5ft D. 50.21ft

For numbers 295-297, refer to the given below.

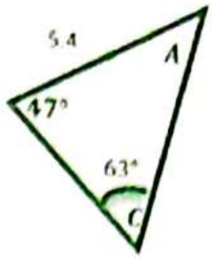
In ΔABC, a = 14 cm, b = 9 cm, c = 6 cm

295. What is the measure of ∠A?
A. 145° C. 137°
B. 167° D. 120°

296. Find the measure of angle B.
A. 23° C. 25°
B. 24° D. 26°

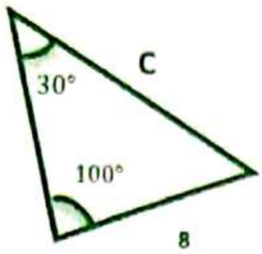
297. What is the measure of ∠C?
A. 16° C. 18°
B. 17° D. 19°

298. Calculate side b.



A. 4.43 B. 4.50 C. 4.91 D. 4.97

299. Calculate side "c".



A. 10.8 B. 11.8 C. 13.8 D. 15.8

300. Law of Sines can be used to solve the following oblique triangle cases **EXCEPT** ____.

A. Side-Angle-Angle
B. Angle-Side-Angle
C. Side-Side-Angle
D. Angle-Angle-Angle

2. Simplify trigonometric expressions using identities

301. Simplify $\frac{\sin^2\theta}{\cos\theta}+\frac{\cos^2\theta}{\cos\theta}$.

A. tanθ
B. sinθ
C. sec θ
D. cot θ

302. Simplify $\frac{tanx}{sinx}$.

A. tanx
B. sinx
C. secx
D. cotx

303. Simplify siny+siny $\cot^2 y$.

A. sec y
B. cos y
C. sin y
D. csc y

304. Simplify cosx+ sinxtanx

A. secx
B. cosx
C. sinx
D. csc x

305. Simplify $\frac{\tan^2 t+1}{1+\cot^2 t}$

A. $\tan^2 t$
B. $\cot^2 t$
C. $\sin^2 t$
D. $\sec^2 t$

306. Simplify $\frac{1-\cos^2 t}{\sin^2 t}$

A. -2
B. 2
C. 1
D. -1

307. Simplify $\frac{\tan^2 t}{1-\sec^2 t}$

A. -2
B. 2
C. 1
D. -1

308. Simplify $\frac{\cos^2 t-1}{\sin^2 t-1}$

A. $\tan^2 t$
B. $\cot^2 t$
C. $\sin^2 t$
D. $\sec^2 t$

309. Simplify $\frac{\sec t \tan t}{\tan^2 t+1}$

A. tan t
B. sin t
C. cot t
D. sec t

310. Simplify $\cot t\,(\tan t + \cot t)$

A. $\csc^2 t$
B. $\sin^2 t$
C. $\tan^2 t$
D. $\cot^2 t$

311. Simplify $\frac{\tan t}{\tan t+\cot t}$

A. $\csc^2 t$
B. $\sin^2 t$
C. $\tan^2 t$
D. $\cot^2 t$

312. Simplify $\cot t \csc t\ (\sec^2 t - 1)$

A. tan t
B. sin t
C. cot t
D. sec t

313. Which of the following identities is $\csc x = \frac{1}{\sin x}$?

A. Fermat's identities
B. Pythagorean identities
C. Quotient identities
D. Reciprocal identities

314. Which of the following identities is $\tan(x) = \frac{\sin x}{\cos x}$?

A. Fermat's identities
B. Pythagorean identities
C. Quotient identities
D. Reciprocal identities

315. Which of the following identities is $\sin^2 u + \cos^2 u = 1$?

A. Fermat's identities
B. Pythagorean Identities
C. Quotient identities
D. Reciprocal identities

3. Describe graphs of the 6 trigonometric functions

For numbers 316-318: Find the amplitude of the given trigonometric functions.

316. $y = 3\sin(x)$

A. 2π
B. 2
C. 3π
D. 3

317. $y = \sin(3x)$

A. 0
B. 1
C. 2
D. 3

318. $y = -2\cos(x)$

A. 0
B. 1
C. 2
D. 3

For numbers 319-321, find the period.

319. $y = 3\sin(x)$

A. 2π
B. 2
C. 3π
D. 3

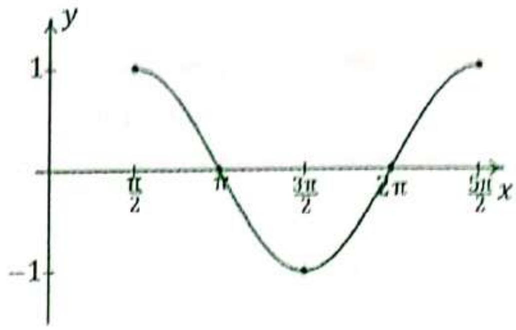
320. $y = \sin(3x)$

A. $\frac{2\pi}{3}$
B. $\frac{3\pi}{2}$
C. 3π
D. 2π

321. $y = -2\cos(x)$

A. $\frac{2\pi}{3}$
B. $\frac{3\pi}{2}$
C. 3π
D. 2π

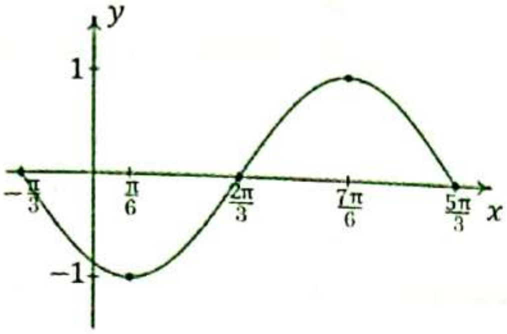
For numbers 322 and 323, refer to the graph below:



322. Which of the following is **TRUE** about the graph?
A. The amplitude is 1.
B. The period is equal to $\frac{3\pi}{2}$.
C. The graph shows a tangent function.
D. The graph has a vertical shift of 2 units.

323. Which of the following shows the equation of the graph?
A. $y = \sin\left(x + \frac{\pi}{4}\right)$
B. $y = \cos(x - \pi)$
C. $y = \sin(x + 3\pi)$
D. $y = \cos\left(x - \frac{\pi}{2}\right)$

For numbers 324-325,refer to the graph below:



324. Which of the following is the equation of the graph?
   A. $y = \cos\left(x - \frac{\pi}{2}\right)$
   B. $y = \sin\left(x + \frac{\pi}{3}\right)$
   C. $y = \cos\left(x - \frac{3\pi}{2}\right)$
   D. $y = \sin\left(x + \frac{\pi}{4}\right)$

325. Which of the following is **INCORRECT**?
   A. The period is 2π.
   B. The amplitude is 1 unit.
   C. The midline is the x-axis.
   D. The phase shift is equal to $\frac{\pi}{2}$.

Probability and Statistics

1. Solve problems involving

a. counting techniques (permutation and combination)

b. probability concepts (addition rule and conditional probability)

c. concepts of binomial and normal distributions

326. If a jar contains 5 red balls and 3 blue balls, how many possible ways are there to get 2 red balls and 1 blue balls?

A. 30 C. 5
B. 27 D. 3

327. Suppose that 4 couples are to be seated in a circular table. If each person must be seated next to his/her partner, how many possible ways are there to seat them in a circular table?

A. 8 C. 24
B. 12 D. 3

328. Which of the following statement is FALSE about the normal distribution?

A. Its curve is bell-shaped.
B. The highest point is at the center of the curve.
C. Mean, median, and mode are equal.
D. The shape of distribution is determined by mean.

329. A student took a five-item multiple choice quiz. If each item has 4 choices, A,B,C, and D, what is the probability that the student's answer has exactly one D?

A. 4 C. 12
B. 405 D. 60

330. In how many ways can the letters from the word *SYZYGY* be arranged?

A. 24 C. 120
B. 240 D. 720

331. In how many possible ways can five item True or False questions be answered?

A. 7 C. 32
B. 25 D. 10

332. Ana is planning to take one English course and two Math courses in college. A list of and 7 Math courses in which she can enrol. In how many possible ways can she choose her desired course?

A. 63 C. 21

B. 10 D. 7

333. Earl plans to go to Bicol to visit his parents. He can reach Bicol by sea, by air, or by land. Suppose that there are 3 ways by sea, 2 ways by air, and 5 ways by land. In how many ways can he go to Bicol?

A. 63 C. 21

B. 10 D. 7

334. There are 5 girls and 7 boys in a gathering. In how many ways can they be arranged in a row so that all boys form a single block?

A. 720 C. 479 001 600

B. 3 628 800 D. 10 080

335. What is the probability that a chosen card from a standard deck is either a queen or ace?

A. 1/4 C. 1/13

B. 3/4 D. 2/13

336. A storage box contains 25 phones and 4 of those phones are defective. What is the probability that 2 of 3 selected phones are defective?

A. 11.43% C. 5.48%

B. 10.04% D. 6.32%

337. The chances that a student in a particular school live in Rizal is 12.4%. What is the probability that in 10 students, at most 3 of them live in Rizal?

A. 97.32% C. 34.56%

B. 56.93% D. 2.68%

338. If a street magician asked you to choose 2 cards from a standard deck, what is the probability that the first card is an ace and the second card is a king?

A. 15.38% C. 21.3%

B. 6.47% D. 7.62%

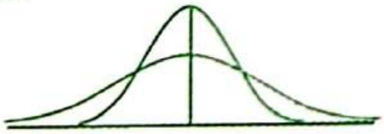
339. There are 2 and 4 possible roads going from city A to city B and from city A to city C, respectively. Going to D, there are 4 and 5 roads from city B and C, respectively. How many possible ways are there to go from A to D?
A. 45
B. 32
C. 16
D. 28

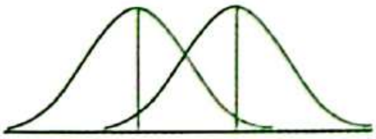
340. A publishing company sent 4, 3, and 5 copies of different textbooks for English, Mathematics, and Science. How many ways are there to arrange the textbooks such that all English books are together?
A. 967 680
B. 161 280
C. 1 560 260
D. 8 709 120

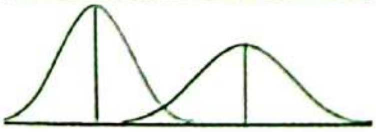
341. Suppose that you roll two dice. What is the probability that the outcome in the first dice is 1 or 3 and the outcome in the second dice is a multiple of 2?
A. 1/4
B. 3/4
C. 1/3
D. 2/3

342. Tito Junie's coin purse consists of five 1-peso coins, three 5-peso coins, and five 10-peso coins. If he gets coins in his purse, in how many ways can he take 18 pesos from his wallet?
A. 363
B. 210
C. 421
D. 720

343. Which of the following illustrates a normal curve with same mean but different standard deviation?

I.


II.


III.


A. I only
B. II only
C. III only
D. I, II, and III

344. According to researchers, there is 35% chance that a random person will answer YES to a question in a survey. Supposed that you interviewed 20 random persons, what is the average number of person that will say YES to the question?

A. 63 C. 21
B. 10 D. 7

345. A jar contains 7 blue balls labeled from 1 to 7, 6 green balls labeled from 1 to 6, and 5 black balls labeled from 1 to 5. What is the probability that 3 selected balls are labeled 2 but different colors?

A. 18 C. 210
B. 12 D. 305

2. Compute/interpret descriptive statistical measures (measures of central tendency and variability)

346. Which of the following is also known as average?

A. Mean C. Variance
B. Median D. Standard Deviation

347. Which is the numerical characteristic of a sample?

A. Mean C. Standard Deviation
B. Parameter D. Statistic

348. What is the standard deviation of a given set of data if its variance is 16?

A. 16 C. 4
B. 8 D. 2

349. What is the mean of the ungrouped data 2, 4, 1, 7, 11?

A. 5 C. 15
B. 10 D. 20

350. Using the same set of data, what is the median?

A. 1 C. 4
B. 2 D. 11

351. Mr. Bisda owns a coffee shop and the following are his gross earnings for the period of 6 months:
Php375 910 Php351 884 Php252 675 Php361 148 Php392740 Php267 900
What is the mean sale of his coffee shop for 6 months?
A. Php325 678.18
B. Php329 079.98
C. Php333 709.50
D. Php331 875.75

352. The average age of 5 people is 42. Another group of people has an average age of 81. If these two groups are combined, what will be the average age?
A. 64
B. 66
C. 68
D. 70

353. Find the median of the set of numbers: 100, 200, 450, 29, 1029, 300, and 2019.
A. 29
B. 300
C. 375.5
D. 450

354. 25 patients were asked to determine the amount of pain they feel while experiencing arthritis on a scale of 1 to 10. Which measure of central tendency would be best to use?
A. Mode, because the data is nominal.
B. Median, because the data is ordinal.
C. Mean, because it is the most used measure of central tendency.
D. Any of the three measures of central tendency is best to use.

355. Three Grade 12 classes have the same mean score on an aptitude test but Class A has a standard deviation three times of Class B and Class C has a standard deviation one-third times of Class B. Which of the three classes is easier to teach?
A. Class A
B. Class B
C. Class C
D. Cannot be determined

3. Make inference on the relationship between two variables from a given correlation coefficient

For numbers 356-360:

The following data shows number of absences and the number of quizzes missed by five students.

STUDENT	1	2	3	4	5
No. of Absences (X)	1	1	2	3	4
No. of Missed Quizzes (Y)	1	2	4	2	4

356. What is the $\sum Y$?
A. 11 B. 13 C. 31 D. 33

357. What is the $\sum X^2$?
A. 11 B. 14 C. 31 D. 33

358. What is the $\sum XY$?
A. 11 B. 14 C. 31 D. 33

359. What is the value of the Pearson r?
A. - 0.43 B. 0.53 C. 0.63 D. - 0.73

360. What is the strength of the correlation?
A. Low
B. Moderately high
C. High
D. Very high

361. The correlation coefficient is used to determine ______________________.
A. the strength of the relationship between the x and y variables
B. the proximity of the relationship between the x and y variables
C. specific value of the x-variable given a specific value of the y-variable
D. specific value of the y-variable given a specific value of the x-variable

362. A correlation of 0.7 was found between time spent studying and percentage on an exam. What is the proportion of variance in exam scores that can be explained by time spent studying?
   A. 0.30
   B. 0.40
   C. 0.49
   D. 0.70

363. If height is independent of average yearly income, what is the predicted correlation between these two variables?
   A. -1
   B. 0
   C. 1
   D. cannot be determined

364. For which of the following Pearson's *r* shows perfect positive correlation?
   A. -1
   B. 0
   C. 0.5
   D. 1

365. What does a value of $r = -0.1$ mean?
   A. The correlation between the variables is negative and strong.
   B. The correlation between the variables is negative but negligible.
   C. The correlation between the variables is negative and moderate.
   D. The correlation between the variables does not exist.

4. Determine the appropriate statistical test of difference (z-test,t-test, F-test) needed in analyzing a given situation

366. George is a researcher and he had come up with a hypothesis that there is no significant difference between the Mathematics scores of Sections Gumamela and Sampaguita in a standardized test. Which statistical test should he use?
   A. Spearman's Rank Correlation
   B. Tukey's Test
   C. Analysis of Variance
   D. *t*-test

367. Which of the following statistical test follows a parametric assumption?
   A. ANOVA
   B. Chi-Square Test
   C. Mann Whitney U test
   D. Wilcoxon Rank sums test

368. Which of the following best describes the statistical test Analysis of Variance (ANOVA)?
A. ANOVA is a test for homogeneity of variance.
B. ANOVA is a test of homogeneity of equal means.
C. ANOVA is a test of difference between two or more means.
D. ANOVA is a test of difference between two or more variances.

369. After performing ANOVA, which of the following post-hoc test should be used so the researcher could pinpoint which group in the sample differs?
A. Bonferroni's Procedure
B. Fisher's LSD
C. Scheffe's Method
D. Tukey's Test

370. Which of the following hypothesis test evaluates how far the observed sample mean deviates, in standard error units, from the hypothesized population mean?
A. ANOVA
B. F-Max test
C. t-test
D. z-test

371. Which statistical test is best use to test the significant difference of the mean of two groups?
A. ANOVA
B. t-test
C. Wilcoxon Rank Sums Test
D. z-test

372. The statistical test, z-test, is appropriate to use when ____________________.
I. the population standard deviation is known
II. the shape of the population is known to be normal
III. the population is relatively large
A. I & II
B. I & III
C. II & III
D. I, II, & III

373. The statistical test, t-test, is appropriate to use when ____________________.
- the population standard deviation is estimated from the sample.
- the shape of the population is known to be normal.
- the population is relatively small.

A. I only
B. II only
C. I & II only
D. II & III only

374. Dane, Edrian, Neil and Sofia are investigating the effect of instruction on the time required to solve a puzzle. Each of the 20 volunteers is given the same puzzle to be solved as rapidly as possible. Subjects are randomly assigned, in equal numbers, to receive two different sets of instructions prior to the task. One group is told that the task is difficuly(X_1), and the other group is told that the task is easy(X_2). The score for each subject reflects the time in minutes required to solve the puzzle. Which statistical test is best to use to test the difference at 0.05 level of significance?

A. Independent Sample t-test
C. One Sample t-test
B. Paired t-test
D. Two Sample t-test

375. Aldwin, Alex, Dale, Danielle, and Vhermen are psychologists testing whether a series of workshops on assertive training increases eye contacts initiated by shy college students in controlled interactions with strangers. A total of 32 subjects are randomly assigned, 8 to a group, to attend either zero, one, two, or three workshop sessions. The results, expressed as the number of eye contacts during a standard observation period. Which F-test must be used?

A. One-Factor with Measures on Different Subjects
B. One-Factor with Measures on the Same Subjects
C. Two-Factor Analysis of Variance
D. F-Max Test

Analytic Geometry

1. Determine the equation of a line relative to given condition

376. Find the equation of a line parallel to $y = 2x + 1$ and passes through $P\ (1,5)$

A. $y = 2x + 3$ B. $y = 2x - 3$ C. $y = 3x + 2$ D. $y = 3x - 2$

377. Find the equation of a line perpendicular to $y = \frac{3}{4}x - 1$ and passes through $P\ (8\ ,-3)$

A. $y = -\frac{4}{3}x + \frac{17}{3}$

C. $y = -\frac{4}{3}x + \frac{23}{3}$

B. $y = -\frac{4}{3}x + \frac{19}{3}$

D. $y = -\frac{4}{3}x + \frac{25}{3}$

378. Write the equation of a line connecting $(-2{,}0)\,and\,(-7{,}7)$ in slope-intercept form.

A. $y = -\frac{7}{5}x - \frac{14}{5}$ B. $y = \frac{7}{5}x - \frac{14}{5}$ C. $y = -\frac{7}{5}x + \frac{14}{5}$ D. $y = \frac{7}{5}x + \frac{14}{5}$

379. Write the equation of a line in slope-intercept form and in standard form.

Parallel to: $y = -\frac{1}{2}x - 6$ and Passes through $P(-1,-5)$

A. $y = -\frac{1}{2}x + \frac{11}{2}$ B. $y = -\frac{1}{2}x + \frac{11}{2}$ C. $y = -\frac{1}{2}x - \frac{11}{2}$ D. $y = -\frac{1}{2}x - \frac{11}{2}$

$x - 2y = 11$ $x + 2y = 11$ $x - 2y = -11$ $x + 2y = -11$

380. Find the equation of the line going through the given pair of points.
$P\ (-5\ ,6\ )\ and\ Q\ (\ 2\ ,-8\ )$

A. $2x + y - 4 = 0$

C. $2x - y + 4 = 0$

B. $2x + y + 4 = 0$

D. $2x - y - 4 = 0$

381. Write an equation of the vertical line through the point $(3\ ,0)$.

A. $x = 3$ B. $y = 3$ C. $x = 0$ D. $y = 0$

382. Find the equation of a horizontal line through the point $(\ -8\ ,6\ )$

A. $x = -8$ B. $y = -8$ C. $x = 6$ D. $y = 6$

383. Find the equation of the line in standard form $Ax + By = C$, with $A > 0$.

Given: through the points $(-4\ ,6\ )\ and\ (9\ ,10)$

A. $4x + 13y = 94$

C. $4x - 13y = -94$

B. $4x - 13y = 94$

D. $4x - 13y = 94$

384. Write the equation of a line parallel to $x = 5$ and passing through the point $(6, -10)$

A. $x = 6$ B. $y = -10$ C. $y = 6x - 10$ D. $x - 10 = 6$

385. Write the equation of the line passing through the midpoint of the line segment defined by the points (2 , 8)and (0 , 4) and perpendicular to the line $-6x + 12y = 10$

A. $2x - y = 8$

C. $-2x + y = 8$

B. $2x + y = 8$

D. $2x - y = -8$

2. Solve problems involving
a. midpoint of a line segment, distance between 2 points, segment division, slopes of lines, distance between a point and a line, and segment division
b. circle, parabola, ellipse and hyperbola

386. Find the distance between the point P(4,3) and Q(9,15)

A. PQ=13

C. PQ=23

B. PQ=18

D. PQ=3

For numbers 387 – 388, refer to the problem below.

If T(7,5), R(4,4) and I(5,-1) are points in a plane.

387. What is the distance between the point R AND I?

A. $3\sqrt{5}$

C. $2\sqrt{10}$

B. $3\sqrt{10}$

D. $2\sqrt{5}$

388. What is the distance between the point T AND I?

A. $4\sqrt{7}$

C. $2\sqrt{10}$

B. $4\sqrt{10}$

D. $3\sqrt{10}$

389. Segment AP has a midpoint of H(3,4) and one the endpoint is A(3,0). What is the coordinates of the other endpoint P?

A. P(8,8)

C. P(0,8)

B. P(8,9)

D. P(9,8)

390. The formula $(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2})$ is called?

A. Distance Formula
B. Midpont Formula
C. Two-point slope Forumla
D. Segment Formula

391. A diameter of a circle has the extreme points (5,9) and (-1,-3). What would be the coordinates of the center?

A. (2,3)
B. (3,3)
C. (5,9)
D. (0,3)

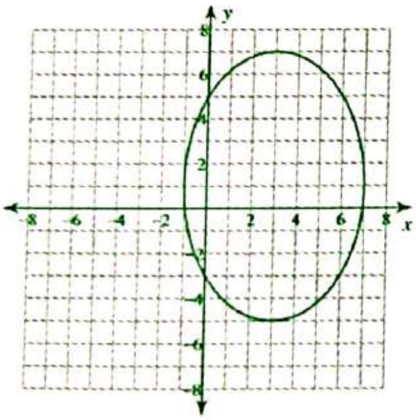
392. Find the coordinates of the midpoint of line segment with an endpoints of C(3.8) and T(7,10).

A. (3,9)
B. (7,5)
C. (5,9)
D. (9,9)

393. Which of the following equations represents a parabola?

A. $x^2+y^2=30$
B. $x^3+y^2=30$
C. $x= y^2$
D. $x^2 = y^2$

For numbers 394 – 395, refer to the figure below.



394. What is the standard form of the given ellipse?

A. $\frac{(x-3)^2}{16}+\frac{(y-1)^2}{36}=1$
B. $\frac{(x-3)^2}{16}+\frac{(y-1)^2}{36}=0$
C. $\frac{(x-3)^2}{15}+\frac{(y-1)^2}{36}=1$
D. $\frac{(x-1)^2}{16}+\frac{(y-3)^2}{36}=1$

395. What is the center of the given ellipse?

A. (2,1)
B. (1,1)
C. (3,1)
D. (3,3)

396. What are the vertices of the given ellipse?
A. (3,7)&(3-5)
B. (3,4)&(3-5)
C. (-5,7)&(3-5)
D. (3,7)&(-3-5)

397. A parabola with a standard equation of $y = \frac{1}{4}x^2$ has vertex of __________.
A. (1, 0)
B. (0, ¼)
C. (0,0)
D. $(\frac{1}{4}, 0)$

398. An ellipse with an equation $\frac{(x-1)^2}{25} + \frac{(y+2)^2}{36} = 1$ has a major axis parallel to __________.
A. x - axis
B. y - axis
C. x and y axes
D. no major axis

399. What type of conic section is the general equation $16x^2-9y^2+64x=89-18y$?
A. Circle
B. Parabola
C. Ellipse
D. Hyperbola

400. Find the ratio in which the line- segment joining the points (5,-4) and (2,3) is divided by the x- axis
A. 4:3
B. 2:3
C. 6:3
D. 5:8

401. Find the coordinate of point P that lies along the directed line segment from A (3,4) to B (6,10) and partitions of the segment in the ratio of 3 : 2
A. P (4.4 , 7.5)
B. P (3.8 , 5.6)
C. P (4.8 , 7.6)
D. P (-4.8 , 7.6)

402. Find the slope of the line that passes through the points (-1,0) and (3,8)
A. 1
B. 2
C. 3
D. 8

403. Which describes the steepness of a line and it can be found using the ratio of rise over run between any two points on the line?
A. Segment
B. Slope
C. Midpoint
D. Vertex

404. Find the slope of the line segment connecting the following points (-1,-1) and (1,3).

A. 5
B. 3
C. 6
D. 2

405. Find the distance between the line and point (2,3).

A. $\sqrt{29}$
B. $\sqrt{19}$
C. $\sqrt{39}$
D. $\sqrt{59}$

406. Find the distance from point $(1, -\frac{1}{2})$ to the line y = 5.

A. $\frac{1}{2}$
B. $\frac{11}{2}$
C. $\frac{11}{3}$
D. $\frac{5}{2}$

407. Find the distance between point $(\frac{1}{2}, \frac{1}{2})$ to the line $x = -\frac{1}{2}$.

A. 7
B. 3
C. 1
D. 4

For 408 to 410, refer to the equation below.

$$x^2+y^2-6x-10y-2=0$$

408. How does the standard equation look like?

A. $(x-5)^2+(y+3)^2=34$
B. $(x+5)^2+(y+3)^2=34$
C. $(x-5)^2-(y+3)^2=34$
D. $(x-5)^2+(y-3)^2=34$

409. What is the coordinate of thecenter?

A. (3,5)
B. (5,5)
C. (-3,5)
D. (3,-5)

410. What is the length of radius?

A. $\sqrt{30}$
B. $\sqrt{34}$
C. $\sqrt{64}$
D. $\sqrt{86}$

3. Determine the equations and graphs of a circle, parabola, ellipse and hyperbola

411. Which of the following is an equation of a circle?

A. $2x^2 + 3y^2 + 2x - 5y + 8 = 0$
C. $2x^2 + 2y^2 + 6x - 5y + 9 = 0$
B. $2x^2 - 5y^2 + 2x - 5y + 8 = 0$
D. $2x^2 - y^2 + 2x - y + 8 = 0$

412. In which quadrant can you find the graph of $x^2 + y^2 + 12x - 16y + 84 = 0$?

A. I B. II C. III D. IV

413. Find the equation of a circle tangent to the x-axis whose center at (8 , 7).

A. $(x - 7 + (y - 8)^2 = 49$
C. $(x - 8)^2 + (y - 7)^2 = 64$
B. $(x - 7)^2 + (y - 8)^2 = 64$
D. $(x - 8)^2 + (y - 7)^2 = 49$

414. Identify the conic section by writing the equation in standard form.

$$x^2 + y^2 - 4x - 8y + 16 = 0$$

A. $\frac{(x-2)^2}{4} + \frac{(y-4)^2}{25} = 1$; *ellipse*
C. $\frac{(x+2)^2}{4} + \frac{(y+4)^2}{25} = 1$; *ellipse*
B. $(x - 2)^2 + (y - 4)^2 = 4$; *circle*
D. $(x + 2)^2 + (y + 4)^2 = 4$; *circle*

415. Find the equation of a parabola with vertex at (5 , 4) and focus at (5 , 7).

A. $(x - 5)^2 = 12\,(y - 7)$
C. $(x - 7)^2 = 12\,(y - 5)$
B. $(x - 5)^2 = -12\,(y - 7)$
D. $(x - 7)^2 = -12\,(y - 5)$

416. Identify the equation of a downward parabola whose vertex is the same as the center of the circle $x^2 + y^2 + 2x + 2y - 2 = 0$ with length of the latus rectum equal to 12 units.

A. $x^2 + 2x - 12y + 13 = 0$
C. $x^2 + 2x + 12y - 13 = 0$
B. $x^2 - 2x + 12y + 13 = 0$
D. $x^2 + 2x + 12y + 13 = 0$

417. Find the standard form of the equation of a parabola whose vertex is at the origin and directrix: x = 1.

A. $y^2 = -4x$ B. $y^2 = 4x$ C. $x^2 = -4y$ D. $x^2 = 4y$

418. Which of the following is a parabola that opens to the right?

A. $x^2 - 2y + 4 = 0$ B. $x^2 + 4y - 8 = 0$ C. $y^2 - 4x + 20 = 0$ D. $y^2 + 5x - 20 = 0$

419. Find the equation of the parabola satisfying the following conditions:

Directrix: $y = 1$ Axis of Symmetry: $x = 5$ Focus is 4 units away from the directrix

A. $x^2 - 7x - 8y + 49 = 0$

B. $x^2 - 2x + 8y + 25 = 0$

C. $x^2 + 10x - 5y + 35 = 0$

D. $x^2 - 10x - 8y + 49 = 0$

420. Which of the following shows a horizontal ellipse?

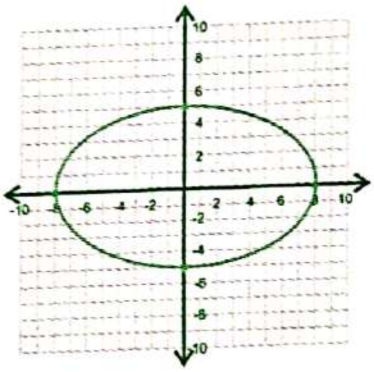
A. $4(x+1)^2 - 9(y-3)^2 = 36$

B. $4(x+2)^2 + 9(y-3)^2 = 36$

C. $9(x+2)^2 - 4(y-1)^2 = 36$

D. $9(x+1)^2 + 4(y-3)^2 = 36$

421. Which of the following is the equation of the given image?



A. $64x^2 + 25y^2 = 1800$

B. $64x^2 - 25y^2 = 1600$

C. $25x^2 + 64y^2 = 1600$

D. $25x^2 - 64y^2 = 1800$

422. Find the ellipse located in quadrant IV.

A. $9x^2 + 4y^2 - 144x + 80y + 940 = 0$

B. $9x^2 + 4y^2 + 144x + 80y + 940 = 0$

C. $9x^2 + 4y^2 - 144x - 80y + 940 = 0$

D. $9x^2 + 4y^2 + 144x - 80y + 940 = 0$

423. Where are the foci of the hyperbola represented by $\frac{x^2}{16} + \frac{y^2}{36} = 1$?

A. $(\pm 4.5, 0)$ B. $(\pm 7.2, 0)$ C. $(0, \pm 4.5)$ D. $(0, \pm 7.2)$

424. Find the equation of the asymptotes represented by the equation $\frac{x^2}{9} - \frac{y^2}{25} = 1$

A. $y = \pm\frac{25}{9}x$ B. $y = \pm\frac{9}{25}x$ C. $y = \pm\frac{3}{5}x$ D. $y = \pm\frac{5}{3}x$

425. If the hyperbola defined by the equation $\frac{(x-4)^2}{1} - \frac{(y-2)^2}{25} = 1$ is translated 5 units down and 3 units to the right, write the equation of the resulting hyperbola.

A. $\frac{(x-7)^2}{1} - \frac{(y-3)^2}{25} = 1$
B. $\frac{(x-7)^2}{25} - \frac{(y+3)^2}{1} = 1$
C. $\frac{(x-7)^2}{1} - \frac{(y+3)^2}{25} = 1$
D. $\frac{(x+7)^2}{1} - \frac{(y+3)^2}{25} = 1$

Calculus

1. Evaluate the limits of given expressions

426. What is the limit of the function $y = x^3 + 3$ as x approaches -1?

A. -1
B. 1
C. -2
D. 2

427. Evaluate $\lim_{x\to-4} \frac{x^2+6x+8}{x+4}$

A. -1
B. -2
C. -3
D. DNE

428. What is the limit of the function $y = x^2 + 4x - 11$ as x approaches 0?

A. -11
B. -10
C. -9
D. -8

429. Evaluate $\lim_{x\to-4} x^2 - 2x + 7$.

A. 29
B. 30
C. 31
D. DNE

430. What is the $\lim_{w\to\frac{1}{3}} 81 - 36w + 9w^2$?

A. 70
B. 75
C. 80
D. 85

2. Determine the derivative of a function

431. The first derivative of $f(x) = -2x^2 + 3x + 5$ is ________.

A. $-2x+3$
B. $4x+3$
C. $-4x+3$
D. $-4x+3+5$

432. Differentiate $y = -4\sqrt{x}$

A. $y' = 2\sqrt{x}$
B. $y' = \frac{2}{\sqrt{x}}$
C. $y' = \frac{4}{\sqrt{x}}$
D. $y' = \frac{-2}{\sqrt{x}}$

433. Differentiate $y = \frac{x^2+3}{5x+4}$.

A. $y' = \frac{5x^2+8x-15}{(5x-4)^2}$
B. $y' = \frac{5x^2+8x-15}{(5x+4)^2}$
C. $y' = \frac{5x^2-8x+15}{(5x+4)^2}$
D. $y' = 5x^2 + 8x - 15$

434. Differentiate $y = 3x\sin(4x)$

A. $y' = 2x\cos(4x) + 3\sin(4x)$
B. $y' = \cos(4x) + \sin(4x) + 3x$
C. $y' = 12x\cos(4x) + 3\sin(4x)$
D. $y' = 2x\cos(4x) + 3\cos(4x^2)$

435. Find the fourth derivative of $y = 2x^5 - 2x^4 + 5x^3 + 6$

A. $y^{(4)} = 10x^4 - 8x^3 + 15x^2$
B. $y^{(4)} = 120x^2 - 48x + 30$
C. $y^{(4)} = 240x - 48$
D. $y^{(4)} = 40x^3 - 24x^2 + 30x$

3. Evaluate Integrals

436. The antiderivative of $\int (x^2 + x + 1)dx$ is

A. $\frac{x^3}{3} + \frac{x^2}{2} + x$
B. $\frac{x^3}{3} + \frac{x^2}{2} + C$
C. $x^3 + x^2 + x + C$
D. $\frac{x^3}{3} + \frac{x^2}{2} + x + C$

437. Evaluate $\int e^{6x-7}dx$

A. $\frac{e^{6x-7}}{6}+C$

B. $e^{6x-7}+C$

C. $-\frac{e^{6x-7}}{6}+C$

D. $-e^{6x-7}+C$

438. Find the antiderivative of $\int \frac{2x-3}{x^2-3x}dx$

A. $\frac{1}{(x^2-3x)^2}+C$

B. $\frac{2x-3}{(x^2-3x)^2}+C$

C. $\ln\left|x^2-3x\right|+C$

D. $\frac{\ln\left|x^2-3x\right|}{2x-3}+C$

439. Find the antiderivative of $\int (7m-2)^2 dm$

A. $\frac{(7m-2)^2}{14}+C$

B. $\frac{(7m-2)^3}{24}+C$

C. $\frac{(7m-2)^3}{3}+C$

D. $\frac{(7m-2)^3}{21}+C$

440. Evaluate $\int -5x sec(4x^2)\tan(4x^2)\,dx$.

A. $\frac{-5\sec(4x^2)}{8}+C$

B. $\frac{5\sec(4x^2)}{8}+C$

C. $-5\sec(4x^2)+C$

D. $\frac{-5\sec(4x^2)}{8}$

4. Solve problems involving differentiation and integration (minimum, maximum, area and volume)

For numbers 441-442, refer to the given function below.

$$f(x)=x^3-3x^2-24x-28$$

441. What is the minimum point of the function?

A. (4 , 108)

B. (-4 , -108)

C. (-4 , 108)

D. (4 , -108)

442. What is the maximum point of the function?

A. (-2 , 0)
B. (-2 , 2)
C. (2 , 0)
D. (2 ,- 2)

For numbers 443 – 444, refer to the problem below.

An airplane is flying towards a radar station at a constant height of 6km above the ground. The distance between the airplane and radar station is decreasing at a rate of 400km per hour when s = 10km. Let x and y be the horizontal and vertical distance respectively and be the displacement.

443. Which of the following equations is the BEST representation for the problem?

A. $y^2 - x^2 = s^2$
B. $s^2 + y^2 = x^2$
C. $x^2 + y^2 = s^2$
D. $x^2 + s^2 = y^2$

444. What is the horizontal speed of the plane?

A. -500 km/hr
B. 500 km/hr
C. -50 km/hr
D. 100 km/hr

445. The volume of spherical glass figurine is increasing at the rate of 8π cubic feet per second. Which of the following equations should be used for the problem?

A. $v = \frac{4}{3}\pi r^2$
B. $v = \frac{1}{3}\pi r^3$
C. $v = \frac{4}{3}\pi r^3$
D. $v = \frac{5}{3}\pi r^3$

446. In reference to problem #445, what is the rate of change when radius is 6 ft?

A. 0.56 ft/sec
B. 0.056 ft/sec
C. 0.0056 ft/sec
D. 5.6ft/sec

For numbers 447 – 448, refer to the problem below.

Suppose a certain car supplies a constant deceleration of A m/s. If it is travelling at 25m/s. when the brakes are applied, its stopping distance is 50m.

447. What is A?

A. 6.35m/s
B. 6.45m/s
C. 6.15m/s
D. 6.25m/s

448. What would have been the stopping distance if the car had been travelling at only 54 kilometers per hour when the brakes were applied?

A. 42m
B. 52m
C. 35m
D. 32m

449. At what speed is the car travelling when the brakes are applied if the stopping distance is 56 meters?

A. 120.17km/hr
B. 110.37km/hr
C. 120.37km/hr
D. 130.37km/hr

450. A spherical balloon is being inflated at the rate of 312in/sec. How fast is the radius r changing at the moment when r=10.20 inches?

A. $\frac{5}{4\pi}$km/hr
B. $\frac{3}{2\pi}$km/hr
C. $\frac{3}{4\pi}$km/hr
D. $\frac{3}{\pi}$km/hr

Modern Geometry, Linear and Abstract Algebra

1. Give characteristics of non-Euclidean geometry which are not found in plane Euclidean geometry

For nos. 451 to 453, choose the letter that satisfies the description for each item

451. A 90-90-90 angle exists in this geometry.

A. Spherical Geometry
B. Hyperbolic Geometry
C. Euclidean Geometry
D. A and B

452. The lines in this geometry is called orthodrome.
A. Spherical Geometry
B. Hyperbolic Geometry
C. Euclidean Geometry
D. A and B

453. It violates Euclid's fifth postulate which state that there is exactly one parallel line that passes through a point P which is not in line l.
A. Spherical Geometry
B. Hyperbolic Geometry
C. Euclidian Geometry
D. A and B

454. The sum of angles of a triangle in hyperbolic geometry is:
A. Exactly 90°
B. More than 90°
C. Not equal to 90°
D. less than 90°

455. If α, β, and γ are the angles of a spherical triangle, the angular excess is given by:
A. $2\pi - (\alpha + \beta + \gamma)$
B. $(\alpha + \beta + \gamma) - 2\pi$
C. $\pi - (\alpha + \beta + \gamma)$
D. $(\alpha + \beta + \gamma) - \pi$

2. Perform operations involving matrices, modular (clock) arithmetic

456. Consider the following matrices below:

$$A = \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix} \qquad B = \begin{bmatrix} 4 & 7 & 3 \\ 2 & 1 & 4 \end{bmatrix} \qquad C = \begin{bmatrix} 1 & 3 & 2 \end{bmatrix}$$

Which of the following matrix multiplication is **NOT** defined?
A. AB
B. BC
C. BA
D. AC

457. Find a, b, c, and d in $\begin{bmatrix} c-d & a+d \\ b+c & c+d \end{bmatrix} = \begin{bmatrix} -5 & 1 \\ 3 & 2 \end{bmatrix}$.
A. $a = \frac{9}{2}, b = -\frac{5}{2}, c = \frac{7}{2}, d = -\frac{3}{2}$
B. $a = -\frac{5}{2}, b = \frac{9}{2}, c = -\frac{3}{2}, d = \frac{7}{2}$
C. $a = \frac{5}{2}, b = -\frac{9}{2}, c = \frac{3}{2}, d = -\frac{7}{2}$
D. $a = -\frac{9}{2}, b = \frac{5}{2}, c = -\frac{7}{2}, d = \frac{3}{2}$

458. What are the entries in the 3rd column of the product of $\begin{bmatrix} 2 & 3 \\ -3 & 7 \end{bmatrix}$ and $\begin{bmatrix} 4 & 2 & 3 \\ 3 & 5 & -1 \end{bmatrix}$?
A. 3 and -16
B. 19 and 29
C. 17 and 9
D. 3 and -1

459. Which of the following matrices is skew-symmetric?

A. $\begin{bmatrix} -2 & -4 & 3 \\ -4 & 3 & 5 \\ 3 & 5 & 4 \end{bmatrix}$

B. $\begin{bmatrix} 1 & 4 & -3 \\ -4 & 7 & -9 \\ 3 & 9 & 8 \end{bmatrix}$

C. $\begin{bmatrix} 5 & 8 & 5 \\ 4 & 7 & 6 \\ 10 & 3 & -3 \end{bmatrix}$

D. $\begin{bmatrix} -2 & 5 & -3 \\ -10 & -3 & -8 \\ 6 & 4 & -8 \end{bmatrix}$

460. Evaluate $3A - 4B + C$ when

$$A = \begin{bmatrix} 1 & 5 & 4 \\ 8 & -3 & 3 \\ 3 & 2 & 2 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 3 & 4 \\ 2 & 6 & 3 \\ -5 & -3 & 2 \end{bmatrix} \quad C = \begin{bmatrix} -3 & 4 & 3 \\ 0 & 0 & 2 \\ 4 & 1 & -5 \end{bmatrix}$$

A. $\begin{bmatrix} 2 & 3 & -3 \\ 19 & 25 & 2 \\ 40 & 19 & -5 \end{bmatrix}$

B. $\begin{bmatrix} -7 & 8 & -9 \\ 61 & 35 & -3 \\ 32 & 19 & -7 \end{bmatrix}$

C. $\begin{bmatrix} -5 & 6 & 4 \\ 19 & 23 & -2 \\ 43 & 18 & -7 \end{bmatrix}$

D. $\begin{bmatrix} -4 & 7 & -1 \\ 16 & 33 & -1 \\ 33 & 19 & -7 \end{bmatrix}$

461. On a 12-hour clock, find x so that $3 + x = 1$.

A. 7
B. 11
C. 10
D. 9

462. What is the multiplicative inverse of 7 under modulo 12?

A. 3
B. 7
C. 6
D. 9

463. Find y such that $7y = 2$ in 12-hour clock.

A. 2
B. 9
C. 4
D. 10

464. What are the entries at the 3rd row in A^T when $A = \begin{bmatrix} 4 & 7 & 5 \\ 3 & 6 & -4 \\ 1 & 3 & -3 \\ -5 & 9 & 2 \end{bmatrix}$

A. 7,6,3,9
B. 5, −4, −3,2
C. 4,3,1 − 5
D. 5,3, −3, −5

465. Which of the following is an example of symmetric matrix?

A. $\begin{bmatrix} -2 & -4 & 3 \\ -4 & 3 & 5 \\ 3 & 5 & 4 \end{bmatrix}$

B. $\begin{bmatrix} 1 & 4 & -3 \\ -4 & 7 & -9 \\ 3 & 9 & 8 \end{bmatrix}$

C. $\begin{bmatrix} 5 & 8 & 5 \\ 4 & 7 & 6 \\ 10 & 3 & -3 \end{bmatrix}$

D. $\begin{bmatrix} -2 & 5 & -3 \\ -10 & -3 & -8 \\ 6 & 4 & -8 \end{bmatrix}$

3. Evaluate 2x2 determinants

466. The determinant of $\begin{bmatrix} 1-b & 3-a \\ 4+a & 2+b \end{bmatrix}$ is given by

A. $b^2 - a^2 + b - a - 10$

B. $b^2 - a^2 + a - b - 10$

C. $a^2 - b^2 + a - b - 10$

D. $a^2 - b^2 + b - a - 10$

467. Compute $\det(A^T)$ if $A = \begin{bmatrix} -2 & 3 \\ 4 & 1 \end{bmatrix}$.

A. 14

B. 1/14

C. -14

D. $-1/14$

468. Find c in $A = \begin{bmatrix} c+3 & -1 \\ 4 & c-2 \end{bmatrix}$ so that the determinant of A is 0.

A. $c = -2, c = 1$

B. $c = 2, c = -1$

C. $c = -3, c = 2$

D. $c = 3, c = -2$

469. If A is 2×2 matrix with $\det(A) = -4$, find $\det(A^3)$.

A. -12

B. 64

C. 12

D. -64

470. Find $\det(A^T B)$ if $\det(A) = 3$and $\det(B^T) = 5$.

A. -15

B. 3/5

C. 5/3

D. 15

471. Which of the following statement about the determinant of matrix is **FALSE**?

I. Determinant is only true for square matrices.
II. Determinant also determines if a matrix is singular.
III. A non-singular matrix has zero determinants.

A. I only
B. II only
C. III only
D. I, II, and III

472. If $A = \begin{bmatrix} 3 & 2 \\ 5 & -1 \end{bmatrix}$ and $B = \begin{bmatrix} 4 & 2 \\ 3 & 1 \end{bmatrix}$. Finddet$(B - 3A)$.

A. 125
B. -170
C. 152
D. -128

473. Which of the following matrices will have a zero determinant?

A. $\begin{bmatrix} 7 & 3 \\ 4 & 5 \end{bmatrix}$
B. $\begin{bmatrix} -2 & 4 \\ 4 & 6 \end{bmatrix}$
C. $\begin{bmatrix} 12 & 9 \\ 4 & 3 \end{bmatrix}$
D. $\begin{bmatrix} -6 & 1 \\ 6 & 1 \end{bmatrix}$

474. Solve for z in $B = \begin{bmatrix} 2 & 3 \\ 1 & z \end{bmatrix}$ if $\det(AB) = 10$and $\det(A) = 2$?

A. -1
B. -5
C. 3
D. 1

475. Which of the following matrices is **NOT ALWAYS** singular?

I. Matrices with at least one row whose entries are all zeroes.
II. Matrices with at least one pair of rows having exactly the same entries.
III. Matrices with at least one row whose entries are all ones.

A. I only
B. II only
C. III only
D. I, II, and III

History of Mathematics, Problem Solving, Mathematical Investigation, Instrumentation and Assessment

1. Identify the pedagogical significance of different tools and devices utilized in the teaching of mathematics

476. What is behaviorism?

A. A theory about teaching, which emphasizes cooperative work and discussion.
B. A theory about teaching which states that teachers are facilitators rather than transmitters of knowledge.
C. A theory about learning, which characterized by the use of stimulus-response situations through which connections are practiced.
D. A theory about learning, which believes that children must be allowed to experiment physically with the things around them if they are to learn.

477. Which of these manipulative materials would help the students discover the formula for the surface area and volume of a prism and pyramid?

A. 3 – D Models
B. 2 – D Models
C. Protractor
D. Geoboards

$A = \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix}$ $B = \begin{bmatrix} 4 & 7 & 3 \\ 2 & 1 & 4 \end{bmatrix}$ $C = \begin{bmatrix} 1 & 3 & 2 \end{bmatrix}$

478. All of these can be performed in excel **EXCEPT** ________________

A. Graph an equation
B. Finds the measures of variation
C. Create tables and worksheets
D. Graph data

479. Which of the following is usually used with the multi-base arithmetic blocks in developing numeration concepts?

A. Fraction mat
B. Number expander
C. Whole-part chart
D. Algebra Tiles

480. What is a good questioning practice?
   A. Asking more authentic questions
   B. Asking more 'how' and 'why' questions
   C. Asking questions answerable by 'yes' or 'no'
   D. Asking questions that require a specific, correct answer

2. Determine the influences of different cultures and mathematicians in the development of different branches of mathematics

481. Who is the First mathematician who attempted to classify curves according to the types of equations that produce them and also made contributions to the theory of equations?
   A. Blaise Pascal
   B. Rene Descartes
   C. Pierre de Fermat
   D. John Napier

482. Who is the Italian mathematician during the renaissance period who was recognized for solving one of the outstanding ancient problems of mathematics, *cubic equations*?
   A. Scipione del Ferro
   B. Regiomontanus
   C. Gerolamo Cardano
   D. Niccolo Tartaglia

483. It is the period where the importance of mathematics in science engendered a new mathematics culture where heavy emphasis was given to the proper instruments for measuring, observing, calculating and recording information.
   A. 18th century
   B. 17th century
   C. Renaissance Period
   D. 16th century

484. What is the Mathematics that is pervasive in identifiable cultural groups?
   A. Conventional mathematics
   B. Ethnomathematics
   C. Socio – mathematics
   D. Cultural mathematics

485. Mathematicians in the Industrial Revolution were engaged in mathematical pursuits either out of pure love for mathematics or for its practical application. By this time many mathematical concepts were already in place. What is this period?
   A. 19th century
   B. 18th century
   C. 17th century
   D. 16th century

3. Cite differences between problem solving and mathematical investigation

486. Which of the following is NOT a part of the core cognitive process?
A. Conjecturing
B. Generalising
C. Imposing
D. Justifying

487. If an individual was given an activity and he is unable to proceed directly to a solution, then that activity is a ______________.
A. bonus
B. problem
C. question
D. task

488. Which of the following statement is **NOT** true about problem solving and mathematical investigation?
A. Problem solving is a task while investigation is an activity.
B. Problem solving is a process of problem posing.
C. The characterization of mathematical investigation does not lie in the open goal of the investigative task itself.
D. Problem solving involves investigation and "other heuristics."

489. Why do problem solving and mathematical investigation?
A. To inform research
B. To pose problems
C. To deviate from routine problems
D. To solve tasks

490. What is the end goal of mathematical investigation?
A. Conjecture
B. Extension
C. Generalization
D. Pattern observation

4. State patterns observed as conjectures

491. Given that the pattern at the right continues, what is the 16^{th} term in the 9^{th} row?

A. 77
B. 78
C. 79
D. 80

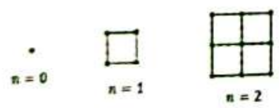
1
2 3 4
5 6 7 8 9
10 11 12 13 14 15 16
⋮

492. Consecutive card towers are built, as shown in the figure at the left. The 1^{st} tower has one floor made of two cards, 2^{nd} tower has two floors made of seven cards and the 3^{rd} tower has 15 cards, and so on. How many cards will the 500^{th} tower have?

A. 250 999
B. 250 500
C. 250 000
D. 255 599

493. Observe the following pattern below. Let X_n be the number of segments that connects an$n$ x n square lattice. Conjecture an expression for X_n.

A. $X_n = n^2 + 1$
B. $X_n = 2n(n+1)$
C. $X_n = 2n(n^2+1)$
D. $X_n = 2n^2(n+1)$



494. Which of the following techniques does NOT prove a conjecture?

A. Direct Proof
B. Counterexample
C. Bijection
D. Inductive Reasoning

495. What is the sum of the 15^{th} row in the Pascal's triangle?

A. 30 768
B. 32 768
C. 32 768
D. 30 768

5. Solve non-routine problems

496. Tickets for a play were sold at Php 200 per child and Php 400 per adult. At one showing, one adult brought 4 children and the remaining adults brought 2 children. The total ticket sales from the children and adults combined is Php 6000. How many children and adult attended the play?

A. 12 children and 9 adults
B. 14 children and 8 adults
C. 16 children and 7 adults
D. 18 children and 6 adults

497. A bookstore charges Php 600 for a yearly membership. The first book is free with the membership, and any book after that costs 76 pesos, including tax. How much money, m, does a student spend after buying b books and a yearly membership?

A. $m = 7.60b$
B. $m = 7.60(b-1)$
C. $m = 7.60b + 60$
D. $m = (7.60b - 1) + 60$

498. There are 18 animals in a pet store. Some are birds and some are dogs. There are 50 legs in all. How many birds are there?

A. 7
B. 8
C. 10
D. 11

499. Mr. Valencia wishes to tile his classroom floor with square tiles. He wants to be able to use whole tiles, without cutting any pieces. The rectangular floor has dimensions 8.4 meters by 7.2 meters. What is the minimum number of whole identical square tiles required and what are the dimensions of each tile?

A. 36 tiles, 1.2mx1.2m
B. 42 tiles, 1.2mx1.2m
C. 36 tiles, 1.4mx1.4m
D. 42 tiles, 1.4x1.4m

500. Kobe and Zildgian were competing in a four-lap, one-mile foot race. Kobe's style was to run at a constant rate of 12 mph. Zildgian decided to run his first three laps at 11 mph, and save a "kick" in order to run a 16 mph pace for the fourth and final lap. Who wins the race, and by how many minutes or seconds?

A. Kobe, by 7 seconds
B. Zildgian, by 7 seconds
C. Kobe, by 1.8 minutes
D. Zildgian, by 1.8 minutes

Arithmetic Number Theory and Business Math

1. C
2. B
3. B
4. D
5. B
6. D
7. A
8. A
9. B
10. D
11. C
12. A
13. C
14. D
15. B
16. B
17. A
18. C
19. D
20. B
21. D
22. A
23. D
24. A
25. C
26. A
27. D
28. C
29. B
30. C
31. D
32. D
33. B
34. C
35. D
36. A
37. D
38. C
39. C
40. A
41. B
42. A
43. A
44. C
45. D
46. B
47. A
48. D
49. B
50. A
51. C
52. B
53. A
54. D
55. A
56. A
57. C
58. C
59. A
60. D
61. B
62. B
63. C
64. B
65. D
66. A
67. B
68. A
69. B
70. B
71. B
72. D
73. C
74. B
75. D

Basic and Advanced Algebra

76. C
77. B
78. D
79. A
80. D
81. D
82. A
83. B
84. B
85. C
86. D
87. D
88. C
89. B
90. D
91. A
92. B
93. C
94. B
95. C
96. A
97. C
98. B
99. A
100. D
101. A
102. D
103. C
104. D
105. B
106. C
107. B
108. C
109. A
110. D
111. C
112. A
113. A
114. D
115. D
116. A
117. B
118. A
119. C
120. D
121. A
122. D
123. A
124. B
125. D
126. D
127. A
128. B
129. A
130. C
131. D
132. B
133. C
134. C
135. A
136. B
137. A
138. D
139. C
140. A
141. C
142. B
143. C
144. B
145. D
146. A
147. A
148. D
149. B
150. C
151. D
152. D
153. B
154. A
155. C
156. C
157. B
158. A
159. C
160. B
161. D
162. A
163. B
164. B
165. C
166. B
167. D
168. C
169. D
170. D
171. D
172. A
173. B
174. B
175. C
176. D
177. B
178. D
179. A
180. D
181. B
182. B
183. D
184. A
185. D
186. A
187. C
188. B
189. D
190. B
191. B
192. D
193. A
194. C
195. B
196. A
197. B
198. B

199. C
200. A

PLANE AND SOLID GEOMETRY

201. D
202. A
203. D
204. D
205. A
206. B
207. D
208. C
209. D
210. B
211. C
212. B
213. D
214. A
215. C
216. C
217. A
218. B
219. C
220. D
221. C
222. B
223. D
224. A
225. B
226. B
227. A
228. C
229. D
230. A
231. D
232. B
233. C
234. C
235. A
236. A
237. D
238. D
239. B
240. C
241. B
242. C
243. A
244. C
245. B
246. B
247. D
248. A
249. A
250. A
251. B
252. B
253. C
254. C
255. A
256. B
257. C
258. B
259. D
260. C
261. A
262. C
263. A
264. D
265. C
266. D
267. B
268. C
269. C
270. A
271. B
272. C
273. B
274. A
275. D

TRIGONOMETRY

276. C
277. B
278. B
279. A
280. B
281. C
282. B
283. C
284. A
285. B
286. A
287. D
288. C
289. A
290. B
291. D
292. C
293. B
294. C
295. C
296. D
297. B
298. A
299. D
300. D
301. A
302. C
303. D
304. A
305. A
306. D
307. C
308. A
309. B
310. A
311. B
312. A
313. B
314. C
315. A
316. D
317. B
318. C
319. A
320. A
321. D
322. A
323. D
324. B
325. D

Probability and Statistics

326. A
327. B
328. D
329. B
330. C
331. C
332. A
333. A
334. B
335. D
336. C
337. A
338. A
339. D
340. D
341. C
342. B
343. A
344. D
345. C
346. A
347. D
348. C
349. A
350. C
351. C
352. B
353. B
354. B
355. C
356. B
357. C
358. D
359. C
360. B
361. A
362. C
363. B
364. D
365. B
366. D
367. A
368. C
369. D
370. D
371. B
372. B
373. A
374. A
375. A

ANALYTIC GEOMETRY

376. A
377. C
378. A
379. D
380. B
381. A
382. D
383. C
384. A
385. B
386. A
387. B
388. B
389. D
390. B

391. A
392. C
393. C
394. A
395. C
396. A
397. C
398. B
399. D
400. A
401. C
402. B
403. B
404. D
405. A
406. B
407. C
408. A
409. A
410. B
411. C
412. B
413. D
414. B
415. A
416. D
417. A
418. C
419. D
420. B
421. C
422. A
423. B
424. D
425. C

CALCULUS

426. C
427. B
428. .A
429. C
430. A
431. C
432. D
433. B
434. C
435. C
436. D
437. A
438. C
439. D
440. C
441. D
442. A
443. C
444. A
445. C
446. B
447. D
448. A
449. C
450. C

Modern Geometry, Linear and Abstract Algebra

451. A
452. A
453. D
454. B
455. C
456. A
457. B
458. A
459. B
460. D
461. C
462. B
463. A
464. B
465. A
466. C
467. C
468. A
469. B
470. D
471. C
472. B
473. C
474. D
475. C

History of Mathematics, Problem Solving, Mathematical Investigation, Instrumentation and Assessment

476. C
477. A
478. A
479. D
480. B
481. B
482. A
483. D
484. B
485. A
486. C
487. B
488. B
489. A
490. C
491. D
492. A
493. B
494. B
495. B
496. A
497. C
498. D
499. B
500. A